

Oracle® Manufacturing Execution System

for Process Manufacturing User's Guide

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Oracle Manufacturing Execution System for Process Manufacturing User's Guide, Release 12.2

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Oracle Manufacturing Execution System for Process Manufacturing User's Guide, Release 12.2 Part No. E49057-15

Oracle welcomes customers' comments and suggestions on the quality and usefulness of this document. Your feedback is important, and helps us to best meet your needs as a user of our products. For example:

- Are the implementation steps correct and complete?
- Did you understand the context of the procedures?
- Did you find any errors in the information?
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If you find any errors or have any other suggestions for improvement, then please tell us your name, the name of the company who has licensed our products, the title and part number of the documentation and the chapter, section, and page number (if available).

Note: Before sending us your comments, you might like to check that you have the latest version of the document and if any concerns are already addressed. To do this, access the new Oracle E-Business Suite Release Online Documentation CD available on My Oracle Support and www.oracle.com. It contains the most current Documentation Library plus all documents revised or released recently.

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Preface

Intended Audience

Welcome to Release 12.2 of the *Oracle Manufacturing Execution System for Process Manufacturing User's Guide*.

This guide assumes that you have working knowledge of your business area's processes, tools, principles, and customary practices. It also assumes that you are familiar with Oracle Process Manufacturing. If you have never used Oracle Process Manufacturing, we suggest you attend one or more of the Oracle Process Manufacturing training classes available through Oracle University.

See Related Information Sources on page xii for more Oracle E-Business Suite product information.

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Related Information Sources

Related Guides

Oracle Process Manufacturing shares business and setup information with other Oracle Applications products. Therefore, you may want to refer to other guides when you set up and use Oracle Process Manufacturing. You can read the guides online in the Oracle Help Center. Navigate to <https://docs.oracle.com>. Select On-Premises Applications, then E-Business.

Guides Related to All Products

Oracle E-Business Suite User's Guide

This guide explains how to navigate, enter and query data, and run concurrent requests using the user interface (UI) of Oracle E-Business Suite. It includes information on setting preferences and customizing the UI. In addition, this guide describes accessibility features and keyboard shortcuts for Oracle E-Business Suite.

Oracle Alert User's Guide

This guide explains how to define periodic and event alerts to monitor the status of your Oracle Applications data.

Oracle E-Business Suite Concepts

This book is intended for all those planning to deploy Oracle E-Business Suite Release 12.2, or contemplating significant changes to a configuration. After describing the Oracle E-Business Suite architecture and technology stack, it focuses on strategic topics, giving a broad outline of the actions needed to achieve a particular goal, plus the installation and configuration choices that may be available.

Oracle E-Business Suite Flexfields Guide

This guide provides flexfields planning, setup, and reference information for the Oracle Applications implementation team, as well as for users responsible for the ongoing maintenance of Oracle Applications product data. This guide also provides information on creating custom reports on flexfields data.

Oracle Application Framework Personalization Guide

This guide covers the design-time and run-time aspects of personalizing applications built with Oracle Application Framework.

Oracle E-Business Suite Installation Guide: Using Rapid Install

This book is intended for use by anyone who is responsible for installing or upgrading Oracle E-Business Suite. It provides instructions for running Rapid Install either to carry out a fresh installation of Oracle E-Business Suite Release 12.2, or as part of an upgrade to Release 12.2.

Oracle Application Server Adapter for Oracle Applications User's Guide

This guide covers the use of OracleAS Adapter in developing integrations between Oracle applications and trading partners.

Oracle e-Commerce Gateway User's Guide

This guide describes the functionality of Oracle e-Commerce Gateway and the necessary setup steps in order for Oracle Applications to conduct business with trading partners through Electronic Data Interchange (EDI). It also contains how to run extract programs for outbound transactions, import programs for inbound transactions, and the relevant reports.

Oracle e-Commerce Gateway Implementation Guide

This guide describes implementation details, highlights additional setups for trading partner, code conversion, and Oracle Applications as well as provides the architecture guidelines for transaction interface files. This guide also contains troubleshooting information and how to customize EDI transactions.

Oracle Report Manager User's Guide

Oracle Report Manager is an online report distribution system that provides a secure and centralized location to produce and manage point-in-time reports. Oracle Report Manager users can be either report producers or report consumers. Use this guide for information on setting up and using Oracle Report Manager.

Oracle iSetup User Guide

This guide describes how to use Oracle iSetup to migrate data between different instances of the Oracle E-Business Suite and generate reports. It also includes configuration information, instance mapping, and seeded templates used for data migration.

Oracle XML Gateway User's Guide

This guide describes Oracle XML Gateway functionality and each component of the Oracle XML Gateway architecture, including Message Designer, Oracle XML Gateway Setup, Execution Engine, Message Queues, and Oracle Transport Agent. The integrations with Oracle Workflow Business Event System and the Business-to-Business transactions are also addressed in this guide.

Oracle XML Publisher Report Designer's Guide

Oracle XML Publisher is a template-based reporting solution that merges XML data with templates in RTF or PDF format to produce a variety of outputs to meet a variety of business needs. Using Microsoft Word or Adobe Acrobat as the design tool, you can create pixel-perfect reports from the Oracle E-Business Suite. Use this guide to design your report layouts.

Oracle XML Publisher Administration and Developer's Guide

Oracle XML Publisher is a template-based reporting solution that merges XML data with templates in RTF or PDF format to produce a variety of outputs to meet a variety of business needs. Outputs include: PDF, HTML, Excel, RTF, and eText (for EDI and EFT transactions). Oracle XML Publisher can be used to generate reports based on existing E-Business Suite report data, or you can use Oracle XML Publisher's data extraction engine to build your own queries. Oracle XML Publisher also provides a robust set of APIs to manage delivery of your reports via e-mail, fax, secure FTP, printer, WebDav, and more. This guide describes how to set up and administer Oracle XML Publisher as well as how to use the Application Programming Interface to build custom solutions.

Oracle interMedia User's Guide and Reference

This user guide and reference provides information about Oracle interMedia. This product enables Oracle9i to store, manage, and retrieve geographic location information, images, audio, video, or other heterogeneous media data in an integrated fashion with other enterprise information. Oracle Trading Community Architecture Data Quality Management uses interMedia indexes to facilitate search and matching.

Oracle Self-Service Web Applications Implementation Guide

This manual contains detailed information about the overview and architecture and setup of Oracle Self-Service Web Applications. It also contains an overview of and procedures for using the Web Applications Dictionary.

Guides Related to This Product

Oracle Process Manufacturing Cost Management User's Guide

The Oracle Process Manufacturing Cost Management application is used by cost accountants to capture and review the manufacturing costs incurred in their process manufacturing businesses. The guide describes how to set up and use this application.

Oracle Process Manufacturing Process Execution User's Guide

The Oracle Process Manufacturing Process Execution application lets you track firm planned orders and production batches from incoming materials through finished goods. Seamlessly integrated to the Product Development application, Process Execution lets you convert firm planned orders to single or multiple production batches, allocate ingredients, record actual ingredient usage, and then complete and close production batches. Production inquiries and preformatted reports help you optimize inventory costs while maintaining a high level of customer satisfaction with on-time delivery of high quality products. The *Oracle Process Manufacturing Process Execution User's Guide* presents overviews of the tasks and responsibilities for the

Production Supervisor and the Production Operator. It provides prerequisite setup in other applications, and details the windows, features, and functionality of the application.

Oracle Process Manufacturing Product Development User's Guide

The Oracle Process Manufacturing Product Development application provides features to manage formula, routing, recipe, and validity rule development within process manufacturing operations. Use it to manage multiple laboratory organizations and support varying product lines throughout the enterprise. Characterize and simulate the technical properties of ingredients and their effects on formula performance and cost. Simulate and optimize formulations before beginning expensive laboratory test batches. Product Development coordinates each development function to provide a rapid, enterprise-wide implementation of new products in your plants. The guide describes how to set up and use this application.

Oracle Process Manufacturing Quality Management User's Guide

The Oracle Process Manufacturing Quality Management application provides features to test material sampled from inventory, production, or receipts from external suppliers. The application lets you enter specifications and control their use throughout the enterprise. Customized workflows and electronic record keeping automate plans for sampling, testing, and result processing. Compare specifications to assist in regrading items, and match customer specifications. Aggregate test results and print statistical assessments on quality certificates. Run stability testing with unrivaled ease. Several preformatted reports and inquiries help manage quality testing and reporting. The guide describes how to set up and use this application.

Oracle Process Manufacturing System Administration User's Guide

This guide provides information on the few tasks that are specific to Oracle Process Manufacturing. It offers information on performing Oracle Process Manufacturing file purge and archive, and maintaining such things as responsibilities, units of measure, and organizations.

Oracle Process Manufacturing Regulatory Management User's Guide

Oracle Process Manufacturing Regulatory Management provides solutions for document management that help meet the FDA 21 CFR Part 11 and other international regulatory compliance requirements. Regulatory information management is facilitated by use of electronic signatures. Manage hazard communications by collaborating with Oracle partners to dispatch safety documents, attached printed documentation sets such as the MSDS to shipments, and set up workflows to manage documentation revisions, approvals, and transmittals. The *Oracle Process Manufacturing Regulatory Management User's Guide* provides the information to set up and use the application.

Oracle Engineering User's Guide

This guide enables your engineers to utilize the features of Oracle Engineering to quickly introduce and manage new designs into production. Specifically, this guide details how to quickly and accurately define the resources, materials and processes necessary to implement changes in product design.

Oracle Inventory User's Guide

This guide describes how to define items and item information, perform receiving and inventory transactions, maintain cost control, plan items, perform cycle counting and physical inventories, and set up Oracle Inventory.

Oracle Bills of Material User's Guide

This guide describes how to create various bills of material to maximize efficiency, improve quality and lower cost for the most sophisticated manufacturing environments. By detailing integrated product structures and processes, flexible product and process definition, and configuration management, this guide enables you to manage product details within and across multiple manufacturing sites.

Oracle Work in Process User's Guide

This guide describes how Oracle Work in Process provides a complete production management system. Specifically this guide describes how discrete, repetitive, assemble-to-order, project, flow, and mixed manufacturing environments are supported.

Oracle Quality User's Guide

This guide describes how Oracle Quality can be used to meet your quality data collection and analysis needs. This guide also explains how Oracle Quality interfaces with other Oracle Manufacturing applications to provide a closed loop quality control system.

Oracle Shipping Execution User's Guide

This guide describes how to set up Oracle Shipping to process and plan your trips, stops and deliveries, ship confirmation, query shipments, determine freight cost and charges to meet your business needs.

Oracle Purchasing User's Guide

This guide describes how to create and approve purchasing documents, including requisitions, different types of purchase orders, quotations, RFQs, and receipts. This guide also describes how to manage your supply base through agreements, sourcing rules and approved supplier lists. In addition, this guide explains how you can automatically create purchasing documents based on business rules through integration with Oracle Workflow technology, which automates many of the key procurement processes.

Other Implementation Documentation

Maintaining Oracle E-Business Suite Documentation Set

Use this guide to help you run the various AD utilities, such as AutoUpgrade, AutoPatch, AD Administration, AD Controller, AD Relink, License Manager, and others. This guide also provides information on maintaining the Oracle applications file system and database.

Oracle Applications User Interface Standards for Forms-Based Products

This guide contains the user interface (UI) standards followed by the Oracle Applications development staff. It describes the UI for the Oracle Applications products and how to apply this UI to the design of an application built by using Oracle Forms.

Oracle Workflow Administrator's Guide

This guide explains how to complete the setup steps necessary for any product that includes workflow-enabled processes. It also describes how to manage workflow processes and business events using Oracle Applications Manager, how to monitor the progress of runtime workflow processes, and how to administer notifications sent to workflow users.

Oracle Workflow Developer's Guide

This guide explains how to define new workflow business processes and customize existing Oracle Applications-embedded workflow processes. It also describes how to define and customize business events and event subscriptions.

Oracle Workflow User's Guide

This guide describes how users can view and respond to workflow notifications and monitor the progress of their workflow processes.

Oracle Workflow API Reference

This guide describes the APIs provided for developers and administrators to access Oracle Workflow.

Oracle eTechnical Reference Manuals

Each eTechnical Reference Manual (eTRM) contains database diagrams and a detailed description of database tables, forms, reports, and programs for a specific Oracle Applications product. This information helps you convert data from your existing applications, integrate Oracle Applications data with non-Oracle applications, and write custom reports for Oracle Applications products. Oracle eTRM is available on My Oracle Support.

Integration Repository

The Oracle Integration Repository is a compilation of information about the service endpoints exposed by the Oracle E-Business Suite of applications. It provides a complete catalog of Oracle E-Business Suite's business service interfaces. The tool lets users easily discover and deploy the appropriate business service interface for integration with any system, application, or business partner.

The Oracle Integration Repository is shipped as part of the Oracle E-Business Suite. As your instance is patched, the repository is automatically updated with content appropriate for the precise revisions of interfaces in your environment.

Do Not Use Database Tools to Modify Oracle E-Business Suite Data

Oracle STRONGLY RECOMMENDS that you never use SQL*Plus, Oracle Data

Browser, database triggers, or any other tool to modify Oracle E-Business Suite data unless otherwise instructed.

Oracle provides powerful tools you can use to create, store, change, retrieve, and maintain information in an Oracle database. But if you use Oracle tools such as SQL*Plus to modify Oracle E-Business Suite data, you risk destroying the integrity of your data and you lose the ability to audit changes to your data.

Because Oracle E-Business Suite tables are interrelated, any change you make using an Oracle E-Business Suite form can update many tables at once. But when you modify Oracle E-Business Suite data using anything other than Oracle E-Business Suite, you may change a row in one table without making corresponding changes in related tables. If your tables get out of synchronization with each other, you risk retrieving erroneous information and you risk unpredictable results throughout Oracle E-Business Suite.

When you use Oracle E-Business Suite to modify your data, Oracle E-Business Suite automatically checks that your changes are valid. Oracle E-Business Suite also keeps track of who changes information. If you enter information into database tables using database tools, you may store invalid information. You also lose the ability to track who has changed your information because SQL*Plus and other database tools do not keep a record of changes.

Understanding MES for Process Manufacturing

This chapter covers the following topics:

- About Oracle MES for Process Manufacturing
- Tasks and Responsibilities
- Process Flow
- Interface to Other Applications

About Oracle MES for Process Manufacturing

Oracle Process Manufacturing closes the gap between conventional Enterprise Resource Planning (ERP) and Manufacturing Execution Systems (MES) for process manufacturers with MES for Process Manufacturing. Batch execution is simplified by paperless record keeping and support for dispensing and pre-weigh operations. Devices are integrated to the manufacturing process. Nonconformances are captured during process execution.

Oracle MES for Process Manufacturing adds additional capabilities to Oracle E-Business Suite applications so that costs, data replication, and risk of error are reduced. Data accuracy, process repeatability, and information visibility are improved by managing all information in a single, real-time database.

MES for Process Manufacturing is designed for use in many process industry plants, especially those looking to move to electronic batch record keeping and those that previously opted to purchase a third-party MES. MES for Process Manufacturing offers the advantage of providing true, enterprise-wide information from the shop floor to the boardroom.

Standard Operating Procedures

Managing Standard Operating Procedures (SOPs) ensures access to the current versions of procedures, thereby reducing the risk of nonstandard manufacturing and quality

practices.

The instruction setup lets you:

- Associate SOPs to a group of instructions, or use them at the unit instruction level
- Specify operator acknowledgement for an instruction
- Specify signatures for an instruction
- Associate a batch action to an instruction
- Associate images or PDF documents to an instruction
- Navigate actions using instructions
- Copy instructions from an existing group
- Define various instruction types for an entity such as clearance, process, dispensing, reverse dispensing

Operator Workbench

The Operator Workbench lets you:

- Identify tasks based on a specified set of search criteria
- Identify batch status at the instruction level
- Lock a task to complete instructions and associated tasks at the batch, operation, activity, and resource level
- View instructions interactively
- Acknowledge instructions
- Navigate to tasks and related actions that use the instructions

Dispensing

Dispensing the proper materials to batches prior to the manufacturing process is a key activity in pharmaceutical, chemical, and food and beverage process industries. The Production process during the manufacture of products in these process industries requires allocation of the right materials with specific quality parameters and precise quantities. Accuracy of these quantities is critical. To enable this, the dispensing functionality is integrated with weigh scale devices.

Dispensing is usually performed in dedicated locations such as environmentally controlled clean rooms that must be maintained in strict accordance with documented

procedures.

Proper labeling of containers that hold dispensed material is also a critical requirement for the positive identification of material and all its related production components.

Device Integration

Device readings are captured directly from the manufacturing equipment during the dispensing process from weighing devices such as scales and balances and as process parameters during batch production.

Device reading capabilities include:

- Synchronous device responses
- Resource-to-device mapping
- Device selection capabilities
- Multiple device interface support using RS232C, USB, RS485 and other common protocols

Nonconformance Tracking

Nonconformance is defined as the difference between the observed conditions and acceptance criteria as stated within an approved protocol such as a master batch record, or if the testing performed deviates from the approved procedure.

Features provided for nonconformance tracking are:

- Nonconformance capture
- Unique identifier for nonconformance
- Classification and disposition of nonconformance
- Corrective actions and preventive actions (CAPA)
- Availability of nonconformance records in the Control Batch Record (CBR)

Label Printing

MES for Process Manufacturing offers the following label printing features:

- Automatic label printing with reprint capabilities
- Configurable label content using XML
- E-signature control

- Label printing history

Master Batch Record

The Master Batch Record (MBR) is the approved and documented set of instructions for manufacturing a specific product, in a specific plant, using specific resources, during a specified date range, to produce a specific batch size. The required elements of MBR are described in the United States Food and Drug Administration (FDA) Code of Federal Regulations (CFR) Title 21 Part 211.186. The recipe created in Oracle Process Manufacturing Product Development application includes material requirements, equipment and resource requirements, instructions, sampling requirements, and notes related to batch manufacturing processing. This recipe is the template that is used repeatedly to manufacture a consistent product.

The three recipe types that can be built in the Oracle Process Manufacturing Product Development application are:

- Master recipes to define how a product is made in a specific process manufacturing cell.
- Site recipes to define how a specific site makes the product, or some portion of the product. A site recipe has the same structure as a general recipe, but may be modified for the local language and unit of measure. It may also be modified to take into account local material availability, or it may only define a part of the general recipe that is actually performed on the site.
- General recipes to define in an equipment-independent manner, the material and process dependencies required to make a product. The general recipe is usually created during or after the pilot plant scaleup of a research and development recipe.

Control Batch Record

The Control Batch Record (CBR) document records all events in the life cycle of a production batch order.

Oracle Manufacturing Execution System (MES) for Process Manufacturing offers the abilities to:

- Control the printing of the MBR and CBR with e-signature control
- Monitor production status
- Modify the format and content of the MBR and CBR
- Monitor external events

Streamlined Electronic Signatures

During the production process, completed tasks are verified on the shop floor using electronic signatures. Oracle E-Records offers a paperless signing process for capturing the requisite electronic signatures on specific events, such as dispensing, reverse dispensing, controlling the progression of business processes and a specific quality result, or creation and modification of key setup data.

Transactions under consideration for approval are listed on the e-signature window. Event signoff status displays on the CBR, and includes the individual who signed, date, time (and time zone), and the reason for signing. Initialing is used to confirm signer identity by entry of an approved user name and password.

The initialing process does not replace the requirement for full signatures on more complex events that require review of information spanning several application windows. Examples of events that require the full signature process include recipe approval, production, batch record closing, and sample results disposition.

Streamlined electronic signatures provide:

- Streamlined navigation flow to complete signatures
- Support for custom approvers and flexible signature rules
- Optionally, use the Oracle Approval Management Engine (AME) to specify users, responsibilities, and the number and hierarchy of signers.

Tasks and Responsibilities

The following is a representative work breakdown structure for managing a manufacturing execution system for process manufacturing within the enterprise. It lists many of the key tasks and the job titles usually responsible for these tasks:

Tasks	Responsibilities
Setting Up Oracle Manufacturing Execution System (MES) for Process Manufacturing	• MES for Process Manufacturing Administrator • Dispensing Operator

Tasks	Responsibilities
Creation and Maintenance of Instructions	• Process Engineer • Formulator • Production Supervisor • MES for Process Manufacturing Administrator
Using the Operator Workbench	• Process Shop Floor Operator • MES for Process Manufacturing Administrator
Dispensing	• Dispensing Operator • MES for Process Manufacturing Administrator
Viewing the Master Batch Record	• MES for Process Manufacturing Administrator • Formulator • Process Engineer
Viewing the Control Batch Record	• Production Supervisor • MES for Process Manufacturing Administrator

Process Flow

The Oracle Manufacturing Execution System (MES) for Process Manufacturing process flow documents the batch life cycle. The business process permits a product designer to define clearance and processing instructions and associate them to a batch, operation, routing, or recipe. By defining individual attributes of these entities and associating them to tasks, the operator can use instructions at runtime to act on the instruction and enter data against it based on its definition. The overall process is controlled by e-signatures that are captured and become part of a permanent record in the Evidence

Store.

1. Define formula, routing, operations, activities and select resources in the Oracle Process Manufacturing (OPM) Product Development application.

Define the instruction sets and instructions.

2. Design a recipe.

Create recipe and define instructions.

3. Create recipe validity rules and review the master batch record.

4. Set up dispensing.

Define dispensable items, dispensing rules, and instructions for the dispensable items.

5. Schedule and release batches.

Reserve the material.

Schedule the shop floor.

Release the batch.

6. Dispense material.

Prewrite designated ingredients.

Dispense material and print labels.

7. Execute batch steps.

Lock steps using the Operator Workbench.

Follow instructions and perform tasks to manage batch operations.

8. Manage quality.

Take samples for testing.

Print quality sample labels.

Log nonconformances.

9. Monitor the batch progress.

Use the interactive batch progression inquiry and report.

Print product labels.

10. Complete the batch.

Complete the tasks.

Resolve and sign off on nonconformances.

11. Validate the control batch record.

Optionally, print the batch record for reference.

12. Close the batch.

Oracle Process Manufacturing (OPM) Product Development is the design source for assembling recipes, formulas, routings, operations, and activities. OPM Process Execution is the production application used to process recipes into batches. Oracle Quality provides the structure for nonconformance tracking. Oracle Warehouse Management System (WMS) provides label printing and device integration infrastructure. Oracle MES for Process Manufacturing uses Oracle XML Publisher to generate its reports and formatted electronic records.

Interface to Other Applications

Oracle Manufacturing Execution System (MES) for Process Manufacturing interfaces with other product applications in the Oracle E-Business Suite to accomplish proper and complete functioning of some of its features. Following are examples of application interfaces:

Oracle Process Manufacturing Process Execution for:

- Defining e-record events to generate the Control Batch Record (CBR)
- Triggering nonconformance tracking
- Acknowledging the instructions
- Material consumption using dispensing

Oracle Process Manufacturing Product Development for:

- Viewing Master Batch Record (MBR)
- Defining instructions

Oracle E-Records for:

- Retrieving XML with e-signature from the E-Records Evidence Store for batch records
- Accepting signers at run time for instructions
- Accepting signers for dispensing setup
- Accepting signers for CBR
- Dispensing and reverse dispensing material

- Printing labels
- Setting the signing rules

Oracle Quality for tracking nonconformances

Oracle Warehouse Management System for:

- Integrating devices
- Printing labels

Oracle XML Publisher for printing reports and electronic records in a human-readable format

Setting Up

This chapter covers the following topics:

- Setting Up Profile Options
- Enabling Oracle MES for Process Manufacturing
- Enabling Control Batch Record
- Setting Up Document Numbering
- Creating and Deleting Document Numbering
- Updating Document Numbering
- Searching for Document Numbering
- Setting Up Dispensing
- Setting Up a Device and Label Printing
- Setting Up Nonconformance Tracking

Setting Up Profile Options

During implementation, set a value for selected profile options to specify how your Oracle MES for Process Manufacturing application controls access to and processes data. The System Administrator sets user profile options at one or more of the following levels: Site, Application, Responsibility, and User.

The following table lists and describes various profile options and their recommended change level.

Profile Option Name	Description	Default	Options	Recommended Change Level
GMO: Control Batch Recording Enabled Date	Enables Control Batch Recording in MES for Process Manufacturing, starting at a particular date and time.	Null. Control Batch Record is disabled.	Date and time Null	Do not manually enter a value. Only the Enable Control Batch Record concurrent program should set this profile option.
GMO: Device Integration Mode	Sets device as enabled or mandated.	Not Applicable.	Auto: Device Read is enabled; manual entry is disabled. Manual: Device Read is disabled; only manual entry is enabled. Both: Device Read is enabled; you can also enter data manually.	Site, Organization
GMO: Device Measure Response Component for Dispense	Identifies the dispensing scale device message component that corresponds to the measurement data element.	Not Applicable.	Provided manually as part of device response setup.	Site, Organization
GMO: Device UOM Response Component for Dispense	Identifies the dispensing scale device message component that corresponds to the unit of measure data element.	Not Applicable.	Provided manually as part of device response setup.	Site, Organization

Profile Option Name	Description	Default	Options	Recommended Change Level
GMO: Device Read Interval (MilliSeconds)	Specifies how often to automatically read data from a material handling device.	Not Applicable.	Provided manually.	Site
GMO: Device Read Timeout (Seconds)	Sets the wait time for the application from an integrated device before returning window control to the user	10	Provided manually.	Site
GMO: Enable Auto Commit for Touch Screen	Enables saving of a mandatory instruction upon Acknowledge without clicking Save.	Not Applicable.	Yes No	Site
GMO: Label Print Mode	Sets label printing mode.	Not Applicable.	Auto: Automatic Label printing. User intervention is not required. Manual: Manual label printing. User intervention is required.	Site
GMO: MES for Process Manufacturing is Enabled	Enables MES for Process Manufacturing	Not Applicable.	Yes: MES for Process Manufacturing is enabled No: MES for Process Manufacturing is disabled (default)	Can only be set to Yes once by the concurrent program.

Profile Option Name	Description	Default	Options	Recommended Change Level
GMO: Validate Step Completion	Warns or stops a step completion in MES if mandatory process instructions are not completed.	Not Applicable.	Stop step completion in MES Warn at step completion in MES	Site
GMO:Operator Certificate for Process Manufacturing is enable	Enables validation of an operator's skills/certification for performing the given action. See: Associating a Certificate or Competency with an Item, Resource or Transaction, page 15-2.	Not Applicable.	Yes No	Site

Profile Option Name	Description	Default	Options	Recommended Change Level
FND:Disable Inline Attachments	Determines whether the Inline Attachments feature is enabled or disabled at the Site or Application level.	True	False: Inline Attachments is enabled. True: Inline Attachments is disabled.	Site, Application Note: The value of the Inline Attachment Enabled property set on the attachment region item overrides this profile option value. Caution: This profile must be set to True to use the MES Operator Workbench shortcut link for viewing batch details. The default is True.

Related Topics

Setting User Profile Options, *Oracle E-Business Suite System Administrator's Guide Documentation Set*

Profile Categories, *Oracle E-Business Suite System Administrator's Guide Documentation Set*

Enabling Oracle MES for Process Manufacturing

Enable Oracle MES for Process Manufacturing by running the Enable MES Functionality concurrent program. This setup is done during application implementation.

To enable the Oracle MES for Process Manufacturing application:

1. Navigate to the MES for Process Manufacturing Administrator responsibility.
2. Choose Run from Request.
3. Select Single Request.
4. Select OK.
5. Select Enable MES for Process Manufacturing Functionality for the request Name.
6. Select OK.
7. Submit the request to run the concurrent program.
8. Verify success by determining that the GMO: MES for Process Manufacturing is Enabled profile option is set to Yes.

Enabling Control Batch Record

Enable Control Batch Record (CBR) by running the Enable Control Batch Record concurrent program. The concurrent program is run before creating batches.

To run the Enable Control Batch Record concurrent program:

1. Navigate to the MES for Process Manufacturing Administrator responsibility.
2. Choose Run from Request.
3. Select Single Request.
4. Select OK.
5. Select Enable Control Batch Record for the request Name.
6. Select OK.
7. Submit the request to run the concurrent program.

Verify success by determining that the GMO: Control Batch Recording Enabled Date profile option lists the enabled date and time.

Setting Up Document Numbering

Document numbers are unique identifiers (IDs) for transactions. Dispensing and the reverse dispensing functions use the document numbering feature to assign a unique

material dispense number and material reverse dispense number.

Assign numbers to the documents manually or automatically. If you use automatic number assignment, then the documents are numbered sequentially. Manual number assignment is useful when you are using preprinted forms or when document numbers are generated by another system.

The document numbering procedure includes:

- Creating document numbering, page 2-7
- Updating document numbering, page 2-8
- Searching for document numbering, page 2-8

Creating and Deleting Document Numbering

Creating document numbering includes selecting a context organization, transaction type, and assignment type.

To create document numbering:

1. Navigate to the Create Document Numbering window.
2. Search and select Dispense or Dispense Reversal as the Transaction.
3. Enter the Organization.
4. Select Assignment Type, choices are:
 - Automatic - Proceed to the next step.
 - Manual - No additional entries are required.
5. Enter Start With as the starting document number. For example, enter 10.
6. Enter the desired Increment between document numbers.
For example, if you enter 1, then the numbering sequence is 1, 2, 3, and so forth. If you enter 10 then the numbering sequence is 10, 20, 30, and so forth.
7. Enter the Length of digits for the document number. Use positive numbers ranging between 1 and 10.
8. Select Zero Padding, choices are:
 - No to display document numbers as 1, 12, and so on.
 - Yes to display document numbers as 01, 0012, and so on.

9. Enter a Prefix.
10. Enter a Suffix.
11. Enter a Separator to display after the prefix and before the suffix. For example, enter #.

The Format Example displays as you enter the values in the different fields. For example, PRE#0000000010#SUF.

Additional Information: For more information about editing document numbering, see Updating Document Numbering, page 2-8.

To delete existing document numbering:

1. Navigate to the Document Numbering window.
2. Select a document number.
3. Click Delete.

Updating Document Numbering

Modify the assignment type, length, zero padding setting, and other values of document numbering.

To update document numbering:

1. Navigate to the Document Numbering window.
2. Search and select a document number.
3. Select Update.
4. Modify the applicable values in the Assignment Type, Zero Padding, Start With, Increment, Length, Prefix, Suffix, and Separator fields.
5. Select Apply.

Searching for Document Numbering

Search for existing document numbers from the Document Numbering window. Update or delete an existing document number.

To search for document numbers:

1. Navigate to the Document Numbering window.
2. Enter any of the following to narrow your search:
 - Search and select an organization by Organization code or Organization Name.
 - Search and select a Transaction type such as Dispense or Dispense Reversal.
3. Select Go. The results of the search appear.

Setting Up Dispensing

Define the dispensing setup to dispense material. Dispensable items in a formula are required to be dispensed prior to consumption when a batch that uses the formula is created. Dispense items that are reservable, recipe-enabled, and active.

The dispensing setup procedure for an item, item and its associated organization, or for a combination of item, organization, and recipe includes:

- Creating a dispensing setup rule
- Updating a dispensing setup rule
- Searching for dispensing setup rule

Creating a Dispensing Setup Rule:

Creating a dispensing setup rule includes indicating a unit of measure (UOM), tolerance type and limits, and adding instructions for the item dispensing operation.

To create a dispensing setup rule:

1. Navigate to the **Create Dispensing Setup** window.
2. Click **Create**.
3. Search for and select an organization by code or organization name.
4. Search for and select a recipe by name or recipe description. If there are several recipe versions, then select a version when you are prompted.
5. Search for and select a recipe-enabled item by name.

Note: Define the dispensing setup for item, item-organization, or a combination of item, organization, and recipe. The combination of

item, organization and recipe is the most restrictive option.

6. Search for and select **Dispense Unit of Measure by Dispense UOM Class**.

Note: The Dispense quantity unit of measure must be convertible to the item's primary UOM. It is independent of other item-assigned UOMs such as those assigned to the secondary quantity, reservation quantity, batch quantity, and formula quantity.

7. Select a tolerance type:

- Percentage
- Quantity

The tolerance limit is applied to the planned quantity and not the reserved quantity. If the dispense quantity is lower than the lower tolerance value, then it is treated as partial dispense.

8. Enter an upper tolerance. Depending on the tolerance type selected, this is either a percentage of the desired dispense quantity or an actual quantity not to exceed.

9. Enter a lower tolerance. Leave this field blank for zero tolerance.

10. Set Dispense Required to Yes to enforce dispensing. The batch enforces dispensing at the batch material line level.

11. Enter start and end dates.

12. Set Security Tag to Yes if you want an alphanumeric tag to label the dispensed material. This security tag is recorded against the dispense in the dispensing window that prints out on the label. The same number is also recorded in batch records, enabling a three-way match of the security code for the seal code, label, and batch record.

13. Select a device integration mode.

You can measure the source container automatically using a device, enter the measurement manually, or use both methods.

- Null (default value). Leave this field blank (null) to use the device integration mode defined in the *GMO: Device Integration Mode* profile option. See: Setting Up Profile Options, page 2-1.
- Auto: Device Read is enabled; manual entry is disabled.

- Both: Device Read is enabled; you can also enter data manually.
 - Manual: Device Read is disabled; only manual entry is enabled.
14. Select a method for creating material transactions.
 - Manual: After dispensing material, manually create a WIP Issue transaction. See: Transacting Materials, *Oracle Process Manufacturing Process Execution User's Guide*.
 - Automatic: After dispensing material, the system automatically creates a WIP Issue transaction.
 15. Enter a dispense expiration fence in hh:mm:ss format. This is the amount of time within which the dispensed material must be used.
 16. To schedule dispensing of material, enter the lead time in the format hh:mm:ss.
 17. Click **Create Instructions** to add an instruction set to the dispensing setup.
 18. Click **Create**.
 19. Enter the instruction set name.
 20. Select one of the following instruction types:
 - Clearance
 - Dispense Clearance
 - Dispense
 - Reverse Dispense
 - Process
 21. Attach an SOP.
 22. Enter an instruction set description.
 23. Click **Apply**.
 24. Obtain electronic signatures for the dispensing setup, if e-record events are enabled.

Note: Enable the oracle.apps.gmo.dispense.setup.create business event to obtain electronic signatures for a dispensing setup rule.

Important: You cannot manually include ingredients that require dispensing in a batch transaction. You must dispense them to a batch.

Updating a Dispensing Setup Rule:

Update a dispensing setup rule from the **Dispensing Setup** window. Modify the tolerance type and tolerance limits and also add instruction sets to the dispensing setup.

To update a dispensing setup rule:

1. Navigate to the **Dispensing Setup** window.
2. Select a dispensing setup from the list of setups available in the search results.
3. Select **Update**.
4. Modify values as needed in the following fields: Dispense Required, Tolerance Type, Upper Tolerance, Lower Tolerance, UOM, End Date, Security Tag Required, Dispense Expiration Fence, and Lead Time.

Note: Once you have dispensed the item, you can only modify the end date.

5. Click **Apply** to only update the dispensing setup rule or click **Update Instructions** to update an instruction set.

Note: If you update an instruction set, you must return to the **Update Dispensing Setup** page and click **Apply** to save the changes to both the dispensing setup rule and the instruction set.

Searching for a Dispensing Setup Rule:

Search for existing dispensing setup rules from the **Dispensing Setup** window. Update or delete an existing setup rule.

To search for dispensing setup rules:

1. Navigate to the **Dispensing Setup** window.
2. Enter any of the following to narrow your search:
 - Search for and select an organization by organization code or organization name.

- Search for and select a recipe by the recipe name or recipe description.
 - Search for and select an item by the item name. The list displays all recipe-enabled items.
 - Search for and select by a status of Not In Use, Past Dues, Used, or All.
3. Click **Go**.
 4. Select **Show Details** to view the dispensing setup details with the associated instructions.

Update an existing setup or manage the dispense instructions.

Setting Up a Device and Label Printing

Following are the steps performed in Oracle MES for Process Manufacturing to use the device integration capability of Oracle Warehouse Management System:

To integrate a mobile device:

1. In the Administrator responsibility, choose Setup.
2. Select Define. Define a new device, or select from the available devices in the list of values.
3. Select Enabled to configure device reading.
4. Set the default Output Method for this device to WMS API.
5. Set Device Type to Scale for dispensing, or to Manufacturing Equipment for process parameters.
6. Specify the subinventory for each device.

For Scales, select the subinventory where dispensing is performed. This is the same selection made for the batch material consumption subinventory. Enter Level as the Subinventory.

7. Assign a message template to the device.
8. Assign the device to the OPM business event—for example, Process Parameter Event or Process Dispensing Event: Choose Assign Devices from Setup.

Additional Information: The organization must be Warehouse Control System-enabled. Device integration is only supported in

OPM for the Process Parameter and Process Dispensing business events.

9. Select Save.

To define request and response directives in Warehouse Management:

1. Request a directive for the OPM business event.

Directives specify the data field sequences for the communication messages between Oracle Applications and the device.

2. Define message templates.

Message templates specify the data field sequences for the communication messages between the device and Oracle Applications.

3. Map device response to the message component codes —for example: Weight UOM Code, Quality, or Batch.
4. Navigate to the Define Devices window and assign the message template defined for the device.

Caution: Automatic Storage and Retrieval System (ASRS) device types can result in an error. Valid options include Scales and Manufacturing Equipment.

To set up label printing:

1. Navigate to Label Printing.
2. Define a printer.
3. Associate a label type to the business flow at the site level. Label types are:
 - Process Dispensing associates to:
 - Dispense Palette
 - Process Dispense Material
 - Process Material
 - Process Production associates to Process Product
 - Process Quality associates to Process Sample

4. Assign a printer to each label type.
5. Create a label format or use the default format shipped.

An XML file is generated with instructions to print a label of specified format to the printer label when you perform a transaction. The XML file is deposited in the specified directory. The third party label printing software retrieves this XML file and prints the label.

Setting Up Nonconformance Tracking

Unexpected events can occur that have direct impact on the manufacturing process and the quality of finished goods. Nonconformances must be tracked and recorded in the batch record and used for analysis by the quality departments for corrective action and preventive action. To document a complete record of a manufacturing batch:

- Evidence of nonconformance must be included in the control batch record
- Details of when the nonconformance occurred
- The phase of the batch in which it occurred
- Details of the incident or observation,
- Level of criticality
- The required follow up and signatures by the operators and supervisors

MES for Process Manufacturing tracks all the manufacturing nonconformances caused by item substitutions, resource substitutions, and dynamic operator observations such as a power interruption or machine failure. Notification of nonconformance may be routed electronically to the reviewers and approvers with the nonconformance details such as electronic signature information logged in the control batch record electronic document.

Copy the seeded nonconformance collection plan to the inventory organizations to log nonconformance events for the production batches.

The following process-specific collection elements must be part of the copied collection plan:

Collection Element	Description
Organization	Process Plant
Batch No	Batch Number

Material	Material
Lot	Lot
UOM	Unit of Measure
Qty	Quantity
Step No	Batch Step Number
Operation	Operation Number with the description associated to it
Activity	Activity Number with the description associated to it
Resource	Resource with the description associated to it
Process Parameter	Resource Process Parameter

When setting up nonconformance tracking based on an operation, it is initiated by specifying the operation number and version, separated by a hyphen (-). For example, if you want the condition to apply to version 1 of operation MIX, enter the following example on the Collection Plan window:

- equals(=) MIX-1

If you want the condition to apply to all or a range of versions, you have to specify a range of all versions; for example:

- between MIX-0 to MIX-99

If you enter equals(=) MIX, the condition will not be met because the batch passes the operation and version separated by a hyphen.

To set up nonconformance tracking in Oracle Quality:

1. Navigate to the Copy Collection Plan window in Oracle Quality. See: Data Collection Using the Quality Workbench, *Oracle Quality User's Guide*
2. Search for the Template Nonconformance Master Plan (Nonconformance only) collection plan. See: Finding a Collection Plan, *Oracle Quality User's Guide*
3. Initialize a copy process by selecting the target organizations.

4. Make all necessary additions or deletions to customize the collection plan.

Note: Go to the Collection Plan window to add or delete collection elements from the collection plan.

5. Enable the Process Manufacturing Transaction.
6. Select Enable Deferred E-Signature to enable the inclusion of the nonconformance in the Control Batch Record.
7. Repeat the process for all organizations where the nonconformance is tracked.

Related Topics

Nonconformance Management, *Oracle Quality User's Guide*

Recording Nonconformances, *Oracle Quality User's Guide*

Instruction Sets and Instructions

This chapter covers the following topics:

- Overview of Instruction Sets and Instructions
- Creating an Instruction Set
- Creating Instruction Attributes
- Creating Instructions
- Adding Instructions to an Instruction Set
- Updating Instructions
- Updating an Instruction Set
- Adding Instructions to an Operation or Recipe

Overview of Instruction Sets and Instructions

Instruction sets are a convenient method to group or categorize instructions. Instruction sets are reusable and the instruction text becomes part of the batch record. Standard Operating Procedures (SOPs) are documents, images, or other files containing information that give detailed instructions on a specific step such as setting up a blender or other plant equipment, and cleaning of equipment. You access a link to an SOP from an individual instruction. An SOP link can be a URL, a document, or another file stored in the Oracle Applications attachments repository. Individual instructions can reference one or more SOPs.

Standard Operating Procedures (SOPs)

SOPs play an important role in assuring quality and adherence to established procedures. It is essential that they are understood and followed by all roles in a manufacturing facility. Create instruction sets and attach SOPs to them to meet your compliance standards.

You can use instructions and instruction sets to attach SOPs to a recipe, to line items of a recipe or for dispensing operations. You can also attach SOPs to a batch during batch

release. Display the SOP during reporting, and provide links to an individual SOP from both the batch and its instructions. Individual instructions can refer to one or more SOPs.

The instruction set is an electronic folder of common instructions. Use instruction sets to group or organize a series of instructions. For example—set up individual instruction sets for pulverizing, blending, mixing, washing, and cooking. These sets serve as containers to help organize groups of instructions with common themes, maintenance requirements, or associations to responsibilities in the work breakdown structure.

Additional Instruction Attributes and Attachments

While following SOPs, you can also capture data related to the SOPs, such as the temperature, humidity, start and end times, and so on. You might also want to attach a file, such as an in-process image of the batch. While you can always attach a file, URL, short text, long text, or comments to any instruction, you must create either clearance or process instruction attributes to capture specific types of data while completing an instruction.

Setting Up Instruction Sets

Instruction sets are the containers for instructions. Use instruction sets to organize, classify, or maintain a series of instructions.

Define an instruction set for any of the following:

- Recipe
- Item in a formula
- Operation, activity in an operation, or resource in an operation
- Operation, activity, resource, or item in a recipe

Additional Information: Override instructions are defined at the lowest level in the hierarchy. The progression from lowest to highest is: item, operation, routing, and recipe. It is recommended that you define instructions at the Operation or Step Quantity level in the recipe. Instructions defined at these levels are automatically presented to the production operator in the Operator Workbench. Instructions defined at other levels, such as Activity, Resource, or Material are available in the Operator Workbench, but the production operator must search for them by selecting a specific entity.

Setting Up Instructions

Instructions are important components of the manufacturing process that provide

details on specific steps. Instructions make processing steps easier to follow. They assure a documented, repeatable process to improve productivity, quality, and consistency throughout the manufacturing process.

Instruction setup streamlines the batch execution process by letting the manufacturing engineers associate viewable characteristics to an instruction. During batch execution, production operators can navigate to the tasks using a single keystroke.

Set up instructions to:

- Drive process execution at the instruction level
- Acknowledge and comment on instructions interactively at the operator level
- Capture attributes and attachments while completing instructions
- Secure e-signature approvals for instructions
- Support different types of instructions
- Associate tasks and Standard Operating Procedures (SOPs) to instructions
- Support instruction at the recipe, process operation, activity, resource, and material level
- Copy instructions from other entities
- Group instructions into reusable instruction sets

Related Topics

[Creating an Instruction Set, page 3-3](#)

[Creating Instructions, page 3-7](#)

Creating an Instruction Set

Create instruction sets and add standard operating procedures (SOPs) to them.

Instruction sets are used to associate several instructions into a logical grouping that can be reused.

Add instructions to the newly created instruction set. Create new instructions or search for and add existing instructions.

You create or access an instruction set from the reference entity. Possible reference entities include:

- Recipe
- Operation

- Routing
- Activity
- Resource
- Formula
- Dispense
- Reverse Dispense

Prerequisites

- Create SOPs.
- Create reference entities such as resources, activities, operations, routings, formula recipe and dispensing setup.

To search for existing instruction sets:

1. Navigate to the Instruction Set window.
2. Select an instruction set.
3. Select Add Existing.
4. Search for an existing instruction set records and select a record.
5. Save your work.

To create an instruction set:

1. Navigate to the Instruction Set window.
2. Select Create.
3. Enter a name for the Instruction Set. Required.
4. Select an Instruction Type.
 - Clearance
 - Dispense Clearance
 - Dispense
 - Reverse Dispense

- Process

Note: You can only select:

- Clearance when creating an instruction set for a recipe.
- Dispense Clearance when creating an instruction set for a dispense area.
- Dispense or Reverse Dispense when creating an instruction set for a dispensing setup.

5. Enter a Description for the instruction set.
6. Select More for SOP to add the related SOPs, or click Add Attachments to add other attachment types.
7. Search for attachments by Title, Description, Category, Last Updated By, Last Update Date, Type, and Usage.
8. Select Go.
9. Enter a Description for the instruction set.
10. Select Apply to save your work.
11. Click Add Attachments to attach an electronic document to the instruction set. The Add Attachment window displays.

Related Topics

[Creating Instructions, page 3-7](#)

[Adding Instructions to an Instruction Set, page 3-9](#)

[Updating an Instruction Set, page 3-10](#)

Creating Instruction Attributes

When creating instructions, you have the option to include custom attributes in the instructions. Examples of custom attributes include:

- Recording the room temperature and humidity while dispensing ingredients.
- Noting the previous batch executed.

- Date of instruction completion.

Before including an attribute in an instruction, you must create the attribute.

Prerequisites

- ☐ Create a lookup type, specifying Manufacturing Execution System for Process Manufacturing as the application, for any instruction attribute that requires a list of values. See: Application Utilities Lookups and Oracle Application Object Library Lookups, *Oracle E-Business Suite Developer's Guide*.

To search for a process instruction attribute:

1. Navigate to the Process Instruction Attributes page.
2. Click Go.

The entire list of attributes appears, alphabetically.

3. In the Attribute Name field, enter a full or partial value.
4. Click Go.

A partial list of attributes, each beginning with the value entered, appears.

To create a process instruction attribute:

1. Navigate to the Process Instruction Attributes page.
2. Click Create.
3. In the Create Process Instruction Attribute page, enter values in the following fields:
 - Code: Enter a unique code.
 - Name: Enter a name for the attribute.
 - Description: Enter a description of the attribute.
 - Type: Select the type of attribute from the following list:
 - Date
 - List Value: If you select List Value, the Lookup code field appears. Search for and select the lookup code which represents the list of values to use with this attribute.
 - Numeric: If you select Numeric, the Min Value and Max Value fields

appear. Enter the minimum and maximum values permitted for this attribute.

- Text

4. Click Apply.

Optionally, click Apply and Create Another to create another process instruction attribute.

To update a process instruction attribute:

You can update an process instruction attribute's Description at any time, but you can only update an attribute's Name and Type fields before adding an attribute to an instruction.

1. After searching for and finding a process instruction attribute, click Update.
2. In the Update Process Instruction Attribute page, update the Name, Description, and Type fields.
3. Click Apply.

To delete a process instruction attribute:

You can only delete a process instruction attribute before the attribute is added to an instruction.

After searching for and finding a process instruction attribute, click Delete.

Creating Instructions

Create instructions, identify the tasks, and add standard operating procedures (SOPs) to the instructions using the Instructions window. You can create instructions for:

- Operation, activity, and resource
- Routing
- Formula
- Recipe
- Item

Tip: Create instructions at the operation level for the most flexibility.

Prerequisites

- Create reference entities such as resources, activities, operations, routings, formula recipe and dispensing setup.
- Create Instructions on the Instruction Create window.
- Determine the structure for the instruction.

To view the list of instructions:

1. Navigate to the Instructions window.
2. Select Instructions to see the existing records.

To create instructions:

1. Navigate to the Create Instruction page by clicking the instruction set name link in the Instruction Set page.
2. Enter a unique value for Instruction Number.
3. Enter Instruction Text description.
4. Select a value for the Acknowledgement your choices are:
 - Optional indicates the acknowledgement is optional for this instruction.
 - Mandatory - Able to Proceed indicates the acknowledgement is mandatory for this instruction. You can proceed to the next instruction and acknowledge it. However, you must return to this instruction to complete it prior to closing the batch or batch step.
 - Mandatory indicates the acknowledgement is mandatory for this instruction. You cannot proceed to the next instruction without acknowledging the current instruction.
5. Enter values in the follow fields:
 - Task: for example, Dispense
 - Task Attribute
 - Task Label, for example, Dispense

Additional Information: Task fields do not appear when defining instructions for an instruction set with a Clearance instruction type.

6. Select More to add the related standard operating procedures.
7. In the Signers tab, specify the Number of Signatures required for validating this instruction and the Role of the signer.

Click Add another row to specify the number of signatures.

Note: A role can be a user name or a responsibility. When a responsibility is used, specify the number of signers who must sign the transaction. When a user name is used, then the number of signers is not valid.

8. Optionally, add instruction attributes. Select the Additional Attributes tab.
9. Click Add Another Row: Additional Attributes.
10. In the Attribute Sequence field, enter an attribute sequence number.
11. In the Attribute Name field, search for and select an instruction attribute.
12. Click Apply or click Apply and Create Another to save the changes and proceed to create another instruction.

Related Topics

Creating an Instruction Set, page 3-3

Creating Instruction Attributes, page 3-5

Adding Instructions to an Instruction Set

To add existing instructions:

1. Navigate to the Add Instructions window.
2. Search for and select an instruction set that has instructions of the same instruction type.
3. Click Next.
4. Select instructions.
5. Click Next.
6. Enter instruction numbers.

7. Click Next.
8. Review selected instructions.
9. Click Finish.
10. Create new instructions or add existing instructions to the set.
11. Move selected instruction up or down the displayed order.

Note: When you copy instructions from an existing instruction set, there is no linkage back to the source instructions. Subsequent changes you make to the source instruction set do not impact instructions copied from this set.

To reorder instructions in the instruction set:

1. Navigate to the Instructions window.
2. View the newly added instructions.
3. Click Reorder.
4. Move selected instruction up or down the list.

Updating Instructions

To update instructions:

1. Navigate to the Update Instructions window.
2. Modify the instruction text, acknowledge, task, task attribute, and the task label information.
3. Click SOP to add more SOPs to the instructions.
4. In the Signers tab, update the number of signers and the role for validating this instruction.
5. In the Additional Attributes tab, update attribute sequences, add more attributes, or delete attributes from the instruction.
6. Click Apply.

Updating an Instruction Set

Use the Update function in the Instruction Set window to modify instruction text, descriptions, and attachments to instructions.

Prerequisites

- Create instruction sets.
- Create instructions.

To update the instruction set:

1. Navigate to the Instructions Set window.
2. Click Update for the instruction set to update.
3. Modify the instruction text and description.
4. Modify Standard Operating Procedure (SOP) attachments.
5. Click Apply.

Related Topics

Creating Instructions, page 3-7

Adding Instructions to an Operation or Recipe

Follow these procedures to add an instruction set and instructions to a recipe or an operation used in a production batch.

To add an instruction set and instructions to a recipe:

1. Navigate to the **Recipe Details** window.
From the Process Engineer responsibility, select **Recipes**.
2. Search for and select a recipe.
3. From the **Actions** menu, select **Process Instructions**.
The **Instruction Set: Recipe Name** page opens.
4. To create instructions for an existing instruction set, go to step 5. Otherwise, create a new instruction set.

To create a new instruction set for a recipe

1. On the **Instruction Set: Recipe Name** page, click **Create**.
2. On the **Create Instruction Set** page, enter values for the following fields:
 - **Instruction Set:** Enter the name of the set.
 - **Instruction Type:** Accept the default value, **Dispense Clearance**.
 - **SOP:** (Optional) Select the **More** link to select a standard operating procedure.
 - **Description:** (Optional) Enter a description of the instruction set.
3. Click **Apply**.

The **Instruction Set: Recipe Name** page now displays the new instruction set for the recipe.

A confirmation message appears: "Instruction Set created successfully."

To create instructions for a recipe's instruction set

5. On the **Instruction Set: Recipe Name** page, click the instruction set name link to create instructions for the set.
6. On the **Instructions** page, click **Create**.

Optional Action: Alternatively, to add an existing instruction to the instruction set, click **Add Existing**.
7. On the **Create Instructions** page, enter values for the following fields:
 - **Instruction Num:** Enter a unique value. Instructions are listed in numerical order.
 - **Instruction Text:** Provide instruction details.
 - **Acknowledgement:** Select one of the following values:
 - **Optional:** Indicates that the acknowledgement is optional for this instruction.
 - **Mandatory - Able to Proceed:** Indicates that the acknowledgement is mandatory for this instruction. You can proceed to the next instruction and acknowledge it. However, you must return to this instruction to complete it prior to closing the batch or batch step.
 - **Mandatory:** Indicates that the acknowledgement is mandatory for this

instruction. You cannot proceed to the next instruction without acknowledging the current instruction.

- **SOP:** (Optional) Select the **More** link to select a standard operating procedure.

To require signatures for the instructions (optional)

8. On the **Signers** tab, click **Add Another Row: Signatures**.
9. Specify the number of signatures required for validating this instruction and the role of the signer.

Note: A role can be a user name or a responsibility. When a responsibility is used, specify the number of signers who must sign the transaction. When a user name is used, then the number of signers is not valid.

To add instruction attributes (optional)

10. On the **Additional Attributes** tab, click **Add Another Row: Additional Attributes**.
11. In the **Attribute Sequence** field, enter an attribute sequence number.
12. In the **Attribute** field, search for and select an instruction attribute.

To finish creating instructions

13. Click **Apply and Create Another** to save the changes and proceed to create another instruction or click **Apply** once you are finished creating instructions.
14. On the **Instructions: Instruction Set** page, click the **Return to Instruction Set: Recipe Name** link.
15. Click **Return** to return to the **Recipe Details** window.

To add an instruction set and instruction to an operation:

1. Navigate to the Process Operation Details:Activities window in the Process Engineer responsibility.
2. Query the desired Operation.
3. Select Process Instructions from the Actions menu. The Instruction Set window displays for the selected operation.
4. If the desired instruction set is displayed, then click it and proceed to step 5.

Otherwise, create the instruction set as follows:

1. Click Create.
2. Enter the Instruction Set name.
3. Select an Instruction Type:
 - Process: Used for production-related tasks that occur after release of the operation step.
 - Clearance: Used for clearance checklist items, such as dispensing or production area cleanliness, calibrations, and ambient conditions, that occur before release of the operation step.

You can create only one instruction set of each instruction type per operation.

4. Click More next to SOP to search for and add an existing SOP.
5. Click Add New to add the SOP as an attachment.
6. Enter a Description for the instruction set.
5. Click Apply. The instruction set is added to the list, and the SOP is attached to the set.
6. Navigate to the Instructions window by clicking the instruction set name link.
7. If the desired instruction is displayed, then click it and proceed to step 8. Otherwise, create the instruction as follows:
 1. Click Create.
 2. Enter Instruction Num as the instruction number. This number is entered automatically when you set up the application for automatic numbering.
 3. Enter the Instruction Text
 4. Select one of the following as Acknowledgement:
 - Optional for an instruction that is informational. Perform this instruction in any order, or skip it. If you do not acknowledge the instruction, then batch processing proceeds without interruption. If you acknowledge the instruction, then sign electronically if the instruction requires an e-signature.
 - Mandatory for an instruction that must be acknowledged and signed electronically if it is configured to require an e-signature. If you do not

acknowledge the instruction, then you cannot proceed to the next instruction. The application requires that all mandatory instructions are acknowledged before the batch step and the batch can be closed.

- **Mandatory:** Able to Proceed for an instruction that must be acknowledged and signed electronically if it is configured to require an e-signature. You can temporarily bypass this instruction and proceed to other instructions, however, the application requires that these instructions are acknowledged before the batch step and the batch can be closed.

5. Search and select a Task.

Note: Task fields do not appear when defining instructions for an instruction set with a Clearance instruction type.

6. Search and select a Task Attribute.

7. Enter a task Label.

8. Select the Signers tab to configure signers for approval:

1. Click Add Another Row in the Signers region.

Conditional Action: You can only add additional signers for the Acknowledgement types Mandatory and Mandatory: Able to Proceed.

2. Add signers using only one of these methods:

- Number of Signatures as 1 and Role as user name.
- Number of Signatures as 1 and Role as responsibility.
- Number of Signatures as the total number of signers required and leave Role blank.
- Number of Signatures as 1 and Role as either user name or responsibility. Enter the lowest ranking role first, followed by any additional roles in increasing rank within the organization.

3. Repeat from step 1 until all signers are added.

9. Select the Additional Attributes tab to add instruction attributes:

1. Click Add Another Row: Additional Attributes.

2. In the Attribute Seq field, enter an attribute sequence number.
 3. In the Attribute field, search for and select an instruction attribute.
10. Click Apply.
 11. Click Reorder to reposition instructions in the list:
 1. Select the instruction.
 2. Click the arrow in the Move Up column to move the instruction up the list.
 3. Click the arrow in the Move Down column to move the instruction down the list.
 4. Click Apply.
 12. Return to the "Note: Navigate to HTML Applications..." message.
 13. Click OK. The process instruction is added to the operation.

Related Topics

Creating Instructions, page 3-7

Operator Workbench

This chapter covers the following topics:

- Requirement
- Understanding the Operator Workbench
- Locking Batch Steps
- Acting on Instructions

Requirement

Following is a requirement that customers have for a simple graphical navigation tool to manage their production processes:

Is there a graphical tool I can use to navigate through production steps?

Yes. The Operator Workbench guides you through a set of predefined instructions created to complete operations required for a production batch. Use the Operator Workbench to navigate and manage the whole production process. This simple tool provides rapid access to every part of your production process.

Operator Workbench Simplifies Navigation

Use the Operator Workbench to identify incomplete tasks. Simple queries help find batches on a wide variety of search criteria such as product, resource, date range, and entity. Simplified, single-click navigation lets you act on an instruction and enter data against it based on the requirement.

Understanding the Operator Workbench

Batch records are work orders that provide a detailed step-by-step course of action to steer production operators through the batch manufacturing process. Depending on individual work instructions, some batch steps can be done at any time, while other steps must be completed before batch production can proceed. Steps can require one or

more signatures to indicate the formal approval to proceed. Entry of production data such as the actual quantity of material added, temperature and pressure observed at various batch steps, or collection of quality samples for in-process testing all add to the complexity of completing the tasks that complete the batch, and must also be archived in the batch record. Oracle Manufacturing Execution System (MES) for Process Manufacturing simplifies this work by putting all the information that the production schedulers, supervisors, and operators need to manage a batch into a single application that virtually eliminates the complexity of paper record keeping. As part of the recipe development process, you can create instructions individually, or in unitized sets that can be organized into reusable procedures. Instructions can be enhanced with images, data files, and paperless attachments that assist in completing batch processing. Associate procedures to the recipe or to major components of the recipe, including its materials, resources, operations, and activities. Clearance instructions can remind or ensure that you clean, clear, or perform checks before starting a batch or operation. Processing instructions can have associated tasks that require specific actions. Actionable tasks and each instruction can require one or more electronic signatures for completion. For example, in the runtime activity for a blending operation, one instruction can require the operator to take the temperature of the blender before material addition. A task is designed to access the process parameter for blender temperature, and another task consumes material into the blender hopper. If either task requires a sign off before the runtime activity can proceed, then an electronic e-signature is mandated. When e-signature permission is granted, the runtime activity can proceed. Both supervisors and production schedulers have top-to-bottom access to real-time production batch status. They can determine batch status immediately using a flexible search engine that displays instruction-level batch details. This provides the agility to take rapid corrective action that is fully documented in the batch audit trail.

The Operator Workbench offers capabilities to:

- Identify tasks to complete. The search window lets you search the batches on a wide variety of search criteria such as product, resource, date range, and entity.
- Lock operations. Take ownership of an operation step by locking it while completing the task. Unlock the step after the task is completed.
- View instructions interactively. View the instructions at material level that are associated to the task such as a recipe, operation, activity, or resource.
- Acknowledge instructions. Acknowledge the instructions and enter comments during the process and indicate to the application that the instruction is completed.
- Navigate to transactional tasks in one click. During batch execution, a single click lets you act on an instruction and enter data against it based on the requirement.

For example, in an instruction to measure the temperature of a liquid and enter it as a resource process parameter, you create the resource process parameter entry as a task that can be accessed directly from its related instruction.

- View the batch summary and navigate to related actions. Navigate to different related actions in an operation, activity, or resource such as quality, resource transaction, and so on.

Operator Workbench Process

Using the Operator Workbench

1. Create the batch.
2. Complete clearance instructions for the batch, if any.
3. Release the batch.
4. Complete step clearance instructions, if any.
5. Release batch steps.
6. Retrieve the worklist of released steps.
7. Lock each step as it will be processed. See: Locking Batch Steps, page 4-3
8. Complete process instructions and related tasks.

Note: Log nonconformances through the OPM Process Execution window interfaces. Refer to the *Oracle Process Manufacturing Process Execution User's Guide* for additional information.

Locking Batch Steps

Use the Operator Workbench to search for and lock a batch step to claim ownership of it.

Prerequisites

- Determine that the desired batch operation or step has a status of WIP.
- Verify your permission to access and lock the batch and its batch steps.

To lock a batch step:

1. Navigate to the Operator Workbench window.
2. Select an Organization code. Required.
3. Click Go.
4. Enter any of the following to narrow your search:

- Search and select the Batch.
 - Search and select the code for the primary Product used in the batch by Product or Description.
 - Search and select the Resource associated with the batch steps by the Resource name or Description.
 - Search and select the Lock Owner by User Display Name or User Name.
5. Select the Status of the batch. Enter any of the following to narrow your search in the context organization entered:
- Select All to find all the available batches listed.
 - Select Closed to find the batches that are closed.
 - Select Complete to find the batches that are completed.
 - Select New to find the new batches.
 - Select Pending to find the batches that are scheduled.
 - Select WIP to find batches that are in process. Oracle Manufacturing Execution System (MES) for Process Manufacturing acts on batches with this status.
 - Select Canceled to find the canceled batches.

Important: The following batch and batch step statuses cannot be processed in the Operator Workbench: New, Pending, Complete, Closed, and Canceled.

6. Select a Planned Start Date from the calendar. It is the planned start date and time of the batch.
7. Select a Planned Completion Date from the calendar. It is the planned completion date and time of the batch.
8. To refine the search, click Show More Search Options, then search and select by name or description the:
- Formula
 - Recipe
 - Routing

- Operation
 - Activity
 - Ingredient
 - Actual Start Date and time of the batch.
 - Actual Completion Date and time of the batch.
9. Click Go.
 10. Select one or multiple batch steps to lock or unlock. The batch steps display in a tabular format. **Status** displays the current batch step statuses as:
 - Locked indicates the batch steps that are locked. You cannot lock the batch steps already locked by others. Lock Owner displays the locked by information.
 - Lock Disabled indicates the batch steps that cannot be locked because the step is not in WIP status.
 - Lock Enabled indicates the batch steps that are available to lock.

Note: You can only lock batch steps that are lock enabled.
 11. Click Lock to lock each batch step.
 12. Take appropriate action on each locked step.
 13. Click Unlock to release batch step ownership.

Acting on Instructions

Use the Operator Workbench to:

- View the instructions.
- Perform actions associated with each instruction.
- Complete instructions and their associated tasks.

Tip: Users with an administrator role who are working on a pending step can also:

- Create new instructions.

- Add existing instructions.
- Add a new instruction set if no instruction set exists.
- Update existing instructions.
- Reorder instructions.

Following is an example of how to act on instructions that are associated with a batch. You can also act on instructions that are associated with other entities.

Prerequisites

- Create instruction sets.
- Create instructions.
- (Optional) Associate tasks with instructions.
- Determine that the desired batch is in the correct status:
 - To execute batch-level clearance instructions, the status must be Pending.
 - To execute step-level clearance instructions, the batch status must be WIP, and the step status must be Pending.
 - To execute process instructions, the batch status must be WIP.
- Verify your permission to access and lock the batch and its batch steps. You must release and lock a batch step before executing process instructions on the step operation.

To act on clearance instructions:

You can only execute clearance instructions for batches and batch steps that are in the Pending status. You must complete mandatory clearance instructions for a batch before releasing the batch. Similarly, after releasing the batch, but before releasing a batch step, you must complete all mandatory clearance instructions for the step.

1. Navigate to the Operator Workbench.
2. Search for a batch.

Tip: If a search results row lists a batch number but no corresponding batch steps, then the batch number has batch clearance instructions. If a search results row lists a batch number

and a step operation, then the batch step operation has batch step instructions. Steps can have both **Clearance** and **Process** instructions, organized under the **Clearance** tab and the **Process** tab.

3. Select the batch number link.

Alternatively, for step instructions, you can select the step operation link.

The clearance instructions for the batch or step operation appear on the **Clearance** tab. For a step operation, any process instructions appear on the **Process** tab.

4. Click **Execute Instructions**.

The first instruction in the instruction set appears.

5. After completing an instruction, select **Instruction Completed**, then click **Continue** to view the next instruction or click **Finish** to stop acting on instructions.

Alternatively, click **Skip** to skip this instruction and view the next instruction.

6. If an electronic signature is required then complete it and click **Close**.

7. Click **Apply** to proceed.

8. Click **Finish**.

9. Perform all pending mandatory instructions and mark them as complete.

10. (Optional) Perform pending optional instructions and mark as them as complete.

Conditional Action: If you choose not to complete every pending optional instruction, you have the option to click **Mark Instruction Set as Complete**. This marks the instruction set with a status of Complete, but gives the incomplete optional instructions a status of Excluded. Once optional instructions are excluded, you no longer have the option to execute them. Alternatively, you can click **Mark All Instructions as Complete**. This marks each optional instruction and the instruction set as complete. This option is not available if one or more of the pending optional instructions requires an e-record and e-signature approval.

To act on process instructions:

1. Navigate to the Operator Workbench.
2. Search for a batch.

Tip: A batch row that is returned without a step listed contains batch clearance instructions. A row that contains a batch number and a step operation contains step instructions. Steps can potentially have both clearance and process instructions, organized under the **Clearance** tab and the **Process** tab.

3. Select the batch number link.

Alternatively, for step instructions, you can select the step operation link.

The clearance instructions for the batch or step operation appear in the **Clearance** tab. For a step operation, process instructions appear in the **Process** tab.

4. Click **Execute Instructions**.

Note: You can only execute process instructions for steps that you locked. If you have not locked a step, then process instructions are read-only.

5. Click **Batch Details**.

6. Review the batch information details such as batch number, batch material, operations, and activities; and the dispense details such as dispense required, dispensed quantity, and WIP plan quantity.

Important: Refer to the *Oracle Process Manufacturing Process Execution User's Guide* for definitions of all fields displayed.

7. Perform actions such as batch update, material update, reservation, nonconformance tracking, and material dispensing.
8. Click **Operation Execution**.
9. Click **Lock**.
10. Display the instruction label.
11. Click **Task** to open the relevant application window.
12. Perform the transaction.
13. Click **Save**.
14. If an electronic signature is required, then complete it and click **Close**.

15. Click **Apply** to proceed.
16. Perform all pending mandatory instructions and mark them as complete.
17. (Optional) If any instruction attributes exist, enter the attribute values.
18. (Optional) Add attachments to instructions.

Additional Information: To include attachments in the Instruction Complete e-record or in the Control Batch Record, perform these pre-requisite steps:

- Verify that you created the batch using a recipe with the Enhanced Instructions attribute.
- Add signers at the instruction level to require e-signatures. See: Updating Instructions, page 3-10.
- Set the business event oracle.apps.gmo.instruction.complete to require e-records and e-signatures.
- For the subscription event oracle.apps.gmo.instruction.complete, set the parameter EDR_COPY_ATTACHMENTS to Y.

If the parameter is set to N (the default value), attachments are not included in the e-record or Control Batch Record.

19. (Optional) Perform all pending optional instructions and mark them as complete.

Conditional Action: If you choose not to complete every pending optional instruction, you have the option to click **Mark Instruction Set as Complete**. This marks the instruction set with a status of Complete, but gives the incomplete optional instructions a status of Excluded. Once optional instructions are excluded, you no longer have the option to execute them. Alternatively, you can click **Mark All Instructions as Complete**. This marks each optional instruction as complete as well as the instruction set as complete. This option is not available if one or more of the pending optional instructions requires an e-record and e-signature approval.

20. Click **Apply**.

Dispensing

This chapter covers the following topics:

- Requirement
- Understanding Dispensing
- Clearing Dispense Areas
- Reserving Material
- Searching for Reserved Material to Dispense
- Executing Dispense Instructions
- Dispensing Reserved Material
- Searching for Dispensed Material
- Viewing the Dispensed Material Activity Details
- Viewing Reverse Dispense Instructions
- Performing Reverse Dispensing
- Viewing the Reverse Dispensed Material Activity Details
- Running Reports for Dispensing Operations

Requirement

Following is a requirement that customers have for supporting dispensing room operations:

Does the application support dispensing room operations?

Yes. Dispensing functionality is a key requirement in the life sciences industry. Dispensing appropriate materials to the right batches prior to the manufacturing process is a key activity in the pharmaceutical and life sciences Industries.

Full Dispensing Room Support

Oracle Manufacturing Execution System (MES) for Process Manufacturing assists the dispensing operator with the in the formulation process by tracking quantities of critical materials used in the manufacturing process. For example, drugs require critical quantities of materials to be added in the manufacturing process. Each material has specific quality parameters and ingredient addition quantities. Oracle Manufacturing Execution System (MES) for Process Manufacturing provides full dispensing room support that is integrated with weigh scale operations to avoid costly manufacturing errors. Drug formulation requires sterile cleanroom environments that must which must be maintained and cleaned according to documented standard operating procedures (SOPs). Weighing equipment must be calibrated and verified. Therefore full dispensing room support for both the environmental conditions and equipment used is essential to accurate drug manufacture.

Understanding Dispensing

Dispensing the right materials to the right batches prior to the manufacturing process is a key activity in life sciences and other process industries. The process is critical when working with potent active pharmaceutical ingredients (APIs) in drug manufacturing. Decimal errors in API calculations can be life threatening. Companies preweigh materials to avoid these errors and to ensure rapid delivery at the critical stages of the drug formulation process.

OPM MES Dispensing lets you deliver material in several modes of operation with full label-printing capabilities. The following functionality is supported:

- Dispensing modes
- Tolerances
- Label printing
- Scale reading
- Reverse dispensing
- Integration to instructions, e-records, and e-signatures
- Dispense Dispatch and Dispense History reports

Oracle Manufacturing Execution System (MES) for Process Manufacturing manages dispensing and preweigh processes to control the labeling, containerization, and delivery of precise quantities of specific materials to their appropriate batch steps. All delivery stages are tracked, including material identification from the point of issue in inventory to point of delivery in the batch manufacturing process. Tracking is critical when working with active pharmaceutical ingredients in drug formulation. Dispensing must meet the guidelines set forth by the U.S. Food and Drug Administration (FDA) regulations (Title 21 Part 211.101). Dispensing is also critical when using ingredients that have hazardous properties, such as the volatile organic compounds (VOCs) used in

paint and plastics manufacturing. Control over the precise delivery of material is essential to ensure a safe manufacturing process and consistent quality product. Standard weighing scale integration into the dispensing process provides a precise, safe, effective delivery of material, while also accounting for any material loss during the transfer of material from one container to another. Material loss may occur due to unavoidable reasons, such as evaporation or material sticking to the container. Oracle Manufacturing Execution System (MES) for Process Manufacturing minimizes the risk of data entry errors by removing error prone manual steps from the batch manufacturing process. Dispense material automatically or manually to labeled pallets in the holding cage. The application generates labels for full traceability throughout the manufacturing process. Oracle Manufacturing Execution System (MES) for Process Manufacturing provides online access to standard operating procedures (SOPs), detailed acknowledgeable work instructions, and simplified e-signature processing. This seamless integration streamlines signoff processes for even the most complex batch operations. Operator and supervisor approvals are orchestrated with a complete audit trail that follows current guidelines for good manufacturing practices.

The dispensing functionality displays dispense pending detail reservations based on the parameters used for query and supports acknowledgement of all instructions defined for dispensing.

The functionality also supports partial dispense. Partial dispense is when the full quantity required is not achieved in one dispense, possibly because of the size of the containers into which you are dispensing. When there is a partial dispense, another dispense is required to complete the allocation of the raw material. For example, the amount required is 112.5 kg, but the minimum number of dispenses to achieve that amount is three: two of 50 kg and one of 12.5 kg. A check is performed on the reserved material before selecting the raw material so that the actual quantity required is displayed in the window.

Approve dispense operations individually using e-records and e-signatures.

Dispensing Process Flow

The overall process to dispense material is the following:

1. Create a batch.
2. Create detailed reservation for material that must be dispensed.
3. Acknowledge the defined instructions.
4. Weigh materials.
5. Perform dispensing.
6. Obtain electronic signatures.
7. Print labels.

Dispensing Features

Following presents the dispensing features provided:

Material Identification

Specify the item to dispense at either the item, item and organization, or item, organization, and recipe level. Identify the material or lots to dispense to the target batch by creating reservations.

Multiple Dispensing Modes

The following dispensing modes are available:

- Full Container Dispense mode is used to deliver the total quantity of material present in the source container. The source container is released to the production batch for consumption.
- Source Container Dispense mode is used to deliver material from a source container. The whole source container is released to the production batch. Required material is dispensed and consumed by the batch. Any remaining material is physically transferred to a new target container, and returned to the warehouse.

For example, a formula requires 45 kilograms (kg) of material. The source container quantity is 50 kg, and 5 kg is removed to a new container in the dispensing process for return to stock. The remaining 45 kg in the source container is dispensed to the batch.

- Target Container Dispense mode is used to deliver material from a target container. Required material is dispensed from a source container to a target container in the dispensing process. The target container material is consumed by the batch. Material remaining in the source container is returned to the warehouse.

Tare Weights

Tare weight is the weight of the actual container when it is empty. The tare weight function lets you:

- Capture tare weight data.
- Store the average tare weight for a container in the item master.
- Use tare weight to calculate the quantity to dispense.

The tare weight of the target container defaults from the entry made in the Oracle Inventory item master. You can enter this value manually.

Note: To set up the container item, navigate to the **Item Master Physical Attributes** tab. Select **Container Item**. The UOM must be the dispense UOM for items dispensed into a container. The weight =

TARE Wt.

Tolerance

Tolerance is the permitted variation between requested or planned dispense quantity and the actual dispensed quantity.

- The dispensing tolerance is stored in the dispensing setup.
- The dispensing tolerance is applied both during dispensing and after confirmation that the dispense is complete. If a batch is only partially dispensed, then the lower tolerance applies only to the total quantity and not to the partial quantity dispensed.
- It is possible to maintain the tolerance either as a percentage or as a quantity.

Label Printing

The label printer automatically prints bar code labels at a specified dispensing location. You can reprint labels. If a label is reprinted, then the print count indicator clearly marks it as a reprint by showing the number of times the label was printed. Printable label types are:

- Dispense Pallet: This label identifies a pallet of dispensed material for a single production batch.
- Process Dispense Material: This label identifies the material dispensed.
- Process Material: This label identifies material left over from a dispensing operation. The label assists in returning the excess material to stores.

You can specify the label to print when you are operating in manual mode. In automatic operation, labels are preselected for printing.

Note: You can select which printer to use and the number of label copies to print when you dispense or reverse dispense material from the **Touchscreen Dispensable Material** and **Touchscreen Dispensed Material Activity** menu options. When you dispense or reverse dispense material from an Oracle Applications Framework page (a non-touchscreen page), one copy of each selected label prints from the default printer.

Reverse Dispensing

A dispensed container can be reverse dispensed in the event of batch cancellation, or when you need to adjust the dispensed material quantity. Reverse dispense either the individual raw material or all the materials for a production batch.

Integration to Instructions

Dispense areas, as well as the dispensing and reverse dispensing pages, are integrated with instructions. Acknowledge instructions associated with a dispense area on the **Dispense Clearance** page. When dispensing or reverse dispensing, acknowledge instructions as you navigate through the **Dispense Instructions** page.

Device or Scale Integration

Dispensing and reverse dispensing windows are integrated with external devices to read weights directly from scales.

E-Record and E-Signature

Dispensing and reverse dispensing windows are integrated with Oracle E-Records to capture e-signatures for dispensing, reverse dispensing, and label printing operations.

Reporting

Generate the following XML-Publisher-based reports to list all the pending dispense operations.

- Pending Dispense Activity List
- Dispense Dispatch

Generate the following XML-Publisher-based report to list all completed dispense operations.

- Completed Dispense
- Dispense History

Clearing Dispense Areas

Your production process may require that you clear the dispense area before starting the dispense process for a batch. Examples of tasks associated with clearing a dispense area include:

- Cleaning
- Calibrating equipment
- Recording dispense booth temperature and humidity conditions

Create dispense clearance instructions for a dispense area to require acknowledgement of these tasks.

To create a dispense clearance instruction set:

1. Navigate to the **Dispense Area** page. Search for an existing dispense area.

Optional Action: Alternatively, you can create a new dispense area with a dispense clearance instruction set. See: Setting up a Dispense Area and Dispensing Booths, page 12-1.

2. Click **Update**.
3. On the **Update Dispense Area** page, in the **Instructions** region, click **Manage Instructions**.
4. On the **Instruction Set: Dispense Area** page, click **Create**.
5. On the **Create Instruction Set** page, enter values for the following fields:
 - **Instruction Set:** Enter the name of the set.
 - **Instruction Type:** Accept the default value, which is Dispense Clearance.
 - **SOP:** (Optional) Select the **More** link to select a standard operating procedure.
 - **Description:** (Optional) Enter a description of the instruction set.
6. Click **Apply**.

A confirmation message appears: "Instruction Set created successfully."

To create dispense clearance instructions:

1. On the **Instruction Set: Dispense Area** page, click the instruction set name link to create instructions for the set.
2. On the **Instructions** page, click **Create**.

Optional Action: Alternatively, to add an existing instruction to the instruction set, click **Add Existing**.

3. On the **Create Instructions** page, enter values for the following fields:
 - **Instruction Num:** Enter a unique value. Instructions are listed in numerical order.
 - **Instruction Text:** Provide instruction details.

- **Acknowledgement:** Select one of the following values:
 - **Optional:** Indicates that the acknowledgement is optional for this instruction.
 - **Mandatory - Able to Proceed:** Indicates that the acknowledgement is mandatory for this instruction. You can proceed to the next instruction and acknowledge it. However, you must return to this instruction to complete it prior to closing the batch or batch step.
 - **Mandatory:** Indicates that the acknowledgement is mandatory for this instruction. You cannot proceed to the next instruction without acknowledging the current instruction.
4. (Optional) To add the related standard operating procedures, select **More**.
 5. (Optional) To require signatures for the instructions:
 1. On the **Signers** tab, click **Add Another Row: Signatures**.
 2. Specify the number of signatures required for validating this instruction and the role of the signer.

Note: A role can be a user name or a responsibility. When a responsibility is used, specify the number of signers who must sign the transaction. When a user name is used, then the number of signers is not valid.
 6. (Optional) To add instruction attributes:
 1. Click **Add Another Row: Additional Attributes**.
 2. In the **Attribute Sequence** field, enter an attribute sequence number.
 3. In the **Attribute** field, search for and select an instruction attribute.
 4. Click **Apply** or click **Apply and Create Another** to save the changes and proceed to create another instruction.

To view a dispense clearance instruction set and instructions for a dispense area:

1. Navigate to the **Dispense Area** page. Search for an existing dispense area.
2. In the **Dispense Area** search results table, select the **Detail** icon for a dispense area.

3. On the **Dispense Area: (Name)** page, in the **Instructions** region, view the associated instruction sets and instructions.

To update a dispense clearance instruction set and instructions:

1. Navigate to the **Dispense Area** page. Search for an existing dispense area.

To update the instruction set

2. In the row for the dispense area, click **Update**.
3. On the **Update Dispense Area** page, in the **Instructions** region, select the instruction set and then click **Manage Instructions**.
4. Click **Update** in the instruction set row.
5. Change the values of the following fields as needed:
 - **Instruction Set**
 - **SOP**
 - **Description**
6. Click **Apply**.

To update one or more instructions

7. On the **Instruction Set** page for the dispense area, click the instruction set name link.
8. On the **Instructions** page, select an instruction and then click **Update**.
9. Change the values of the following fields as needed:
 - **Instruction Num**
 - **Instruction Text**
 - **Acknowledgement**
 - **SOP**
10. On the **Signers** tab, add, change, or delete the number of signatures and roles as needed.
11. On the **Additional Attributes** tab, add, change, or delete the attribute sequences and attribute names as needed.

12. Click **Apply**.

To execute dispense clearance instructions:

1. Navigate to the **Dispense Material** page.
2. Select the organization and dispense area for the batch, and then click **Go**.
3. Enter search criteria for batches that need items dispensed in the selected dispense area, then click **Go**.

The batches requiring items that meet the search criteria appear in the table below.

4. Select a pending dispense item, and then click **Dispense Clearance**.

The **Dispense Clearance Instructions** page for the item opens.

5. After completing the instructions, select **Instruction Completed**.

6. Click **Continue**, if this is an option. Otherwise, click **Finish**.

Clicking **Continue** displays the next instruction to complete. Click **Finish** once all clearance instructions are complete.

A message informs you that all clearance instructions are complete and you are returned to the **Dispensable Material** page.

Related Topics

Overview of Instruction Sets and Instructions, page 3-1

Setting up a Dispense Area and Dispensing Booths, page 12-1

Clearing a Dispense Area Using the Touch Screen Interface, page 12-2

Reserving Material

You must detail reserve material before dispensing it. Refer to the *Oracle Process Manufacturing Process Execution User's Guide* for details on reserving the material.

Refer to "Dispensing Reserved Material" in this guide for detailed information on dispensing the reserved material.

Searching for Reserved Material to Dispense

Following is the procedure to search for reserved material to dispense.

To search for reserved material to dispense:

1. Navigate to the **Dispensable Material** window.
2. Enter any of the following to narrow your search for reserved material.
 - Select an **Organization**. Required. The list displays only the process-enabled organizations.
 - Search and select a **Dispense Area** by the **Description**.
 - Search and select a **Batch**. The list displays only the pending and work in process (WIP) batches in the selected organization.
 - Select a **Batch Step** number.
 - Search and select an **Ingredient** by **Name** or **Description**.
 - Search and select a **Product** by **Name** or **Description**.
 - Select a **From Date** and **To Date** from the calendar to refine the search to the batch material requirement date.
3. Deselect the **Show Partially Dispensed** indicator to display all dispenses where the total dispensed quantity is below the lower tolerance. This is the default. Select this indicator to display all dispenses less than target quantity.
4. Click **Go**.

Executing Dispense Instructions

Use the following procedure to execute material dispense instructions.

To execute material dispense instructions:

1. Search for a reserved material from the **Dispensable Material** window. Refer to Searching for Reserved Material to Dispense, page 5-10 for detailed information.
2. Select an item to dispense.
3. Click **Dispense**.
4. Perform tasks as defined in the instructions. Refer to Acting on Instructions, page 4-5 for detailed information.
5. Click **Finish** to navigate to the **Dispense Material** window.

Dispensing Reserved Material

Following is the procedure to dispense reserved material for each of the listed modes of operation.

To dispense reserved material using the Full Container mode:

1. Navigate to the **Material Dispense** window.
2. Enter the **Dispense Number**. This is the dispense document number. It is valid only if you have configured manual document number entry. Required.
3. Select **Booth** for the chosen Dispense Area.
4. Select dispensing **Mode** as **Full Container**.
5. Search for and select the **Source Scale**.

A box opens, displaying the available scales. Use the **Filter By** option to narrow the list of scales by:

- Dispense Area
 - Booth. Select Booth to view only the scales linked to the booth selected in the preceding step.
6. Search for and select the **Source Container** by **Container Name** or **Description**.
Additional Information: Container information is set up in the Oracle Inventory Item Master.
 7. Click **Read Device** to measure the container weight. The application calculates the full container quantity.
 8. Enter a **Security Tag** number.
 9. Select **Dispense Label** and **Material Label** to indicate label formats for printing.

Material Dispense Page Using the Full Container Mode

Material Dispense: C100001, Maiz Starch(Pharmaceutical RM, PR1 PR1)

Quantity
 Pending Quantity 50 KGM
 Dispensed Quantity **38 KGM**

Selection
 Booth PHARMA Dry Goods
 Mode Full Container
 Include Tare ☒

Context
 Batch 254
 Lot 456
 Hazard No
 Hazard Class

Source
 •Scale SCALE2
 •Container CONTAINER-SD1
 Tare 2 KGM
 •Quantity 40 KGM

Security
 Tag

Label
 ...

Select	Label Type	Printer	Label Count
☐	Dispense		1
☒	Material	LabelPDF	1
☐	Pallet		1

Cancel Continue

10. View the dispense instructions.
11. View the **Partial Dispense History** report.
12. Obtain e-signatures.
13. Print the labels.

To dispense reserved material using the Source Container mode:

1. Navigate to the **Dispense Material** window.
2. Enter the **Dispense Number**. This is the dispense document number. It is valid only if you have configured manual document number entry. Required.
3. Select **Booth** for the chosen Dispense Area.
4. Select dispensing **Mode** as **Source Container**.
5. Search for and select the **Source Scale**.

A box opens, displaying the available scales. Use the **Filter By** option to narrow the list of scales by:

- Dispense Area
- Booth. Select Booth to view only the scales linked to the booth selected in the

preceding step.

6. Search and select the **Source Container** by **Container Name** or **Description**.
7. Search for and select the **Target Scale**, similarly to how you selected the source scale, using the same filtering method.
8. Search and select the **Target Container** by **Container Name** or **Description**.

Note: Click **Read Device** to capture readings directly from an integrated device.

9. The empty target container weight defaults. Modify this weight if required.
10. Measure or enter the source container weight.
11. Transfer the excess material into target container.
12. Measure the target container weight with the material transferred. The dispensed quantity is calculated and displayed. This quantity is the tare adjusted source container weight minus the tare adjusted target container weight.
13. **Re-measure Source Container** weight to verify the dispensed quantity.

Additional Information: If the source container weighs less than expected due to unavoidable loss of material, the quantity discrepancy displays in **Material Loss**. A Miscellaneous Issue transaction is automatically created to account for the material loss. The dispensed quantity is recalculated as the tare adjusted **Re-Measure Source Container** weight.

14. Enter a **Security Tag** number.
15. Select **Dispense Label** and **Material Label** to print labels.

Material Dispense Page Using the Source Container Mode

Material Dispense: C100001, Maiz Starch(Pharmaceutical RM, PR1 PR1)

Quantity
Pending Quantity 50 KGM
Dispensed Quantity 38 KGM

Selection
Booth PHARMA Dry Goods
Mode Source Container
Include Tare ☒

Context
Batch 254
Lot 456
Hazard No
Hazard Class

Source
Scale SCALE2
Container CONTAINER-SD1
Tare 2 KGM
Quantity 40 KGM

Target
Scale Scale3
Container CONTAINER-SD2
Tare 2 KGM
Quantity 2 KGM

Reweigh Source
Quantity 40 KGM

Material Loss
Quantity 0 KGM

Security
Tag

Label

Select	Label Type	Printer	Label Count
☐	Dispense		1
☒	Material	LabelPDF	1
☐	Pallet		1

Cancel Continue

16. View the dispense instructions.
17. View the **Partial Dispense History** report.
18. Obtain e-signatures.
19. Print the labels.

To dispense reserved material using the Target Container mode:

1. Navigate to the **Dispense Material** window.
2. Enter the **Dispense Number**. This is the dispense document number. It is valid only if you have configured manual document number entry. Required.
3. Select **Booth** for the chosen Dispense Area.
4. Select dispensing **Mode** as **Target Container**.
5. Search for and select the **Source Scale**.

A box opens, displaying the available scales. Use the **Filter By** option to narrow the list of scales by:

- Dispense Area

- Booth. Select Booth to view only the scales linked to the booth selected in the preceding step.
6. Search and select the **Source Container** by **Container Name** or **Description**.
 7. Search for and select the **Target Scale**, similarly to how you selected the source scale, using the same filtering method.
 8. Search and select the **Target Container** by **Container Name** or **Description**.

Note: Click **Read Device** to capture readings directly from an integrated device.
 9. The empty target container weight defaults. Modify this weight if required.
 10. Measure or enter the source container weight.
 11. Transfer the required material into target container.
 12. Measure the target container weight with the material transferred. The dispensed quantity is calculated and displayed as the tare adjusted target container weight.
 13. **Re-measure Source Container** weight to verify the dispensed quantity.

Additional Information: If the source container weighs less than expected due to unavoidable loss of material, the quantity discrepancy displays in **Material Loss**. A Miscellaneous Issue transaction is automatically created to account for the material loss.
 14. Enter a **Security Tag** number.
 15. Select **Dispense Label** and **Material Label** to print labels.

Material Dispense Page Using the Target Container Mode

Quantity
Pending Quantity 50 KGM
Dispensed Quantity 38 KGM

Selection
Booth PHARMA Dry Goods
Mode Target Container
Include Tare ON

Context
Batch 254
Lot 456
Hazard No
Hazard Class

Source
Scale SCALE2
Container CONTAINER-SD1
Tare 2 KGM
Quantity 40 KGM

Target
Scale Scale3
Container CONTAINER-SD2
Tare 2 KGM
Quantity 40 KGM

Reweigh Source
Quantity 40 KGM

Material Loss
Quantity 0 KGM

Security
Tag

Label

Select	Label Type	Printer	Label Count
☐	Dispense		1
☐	Material	LabelPDF	1
☒	Pallet		1

Cancel Continue

16. View the dispense instructions.
17. View the **Partial Dispense History** report.
18. Obtain e-signatures.
19. Print the labels.

Searching for Dispensed Material

Use the Dispensed Material Activity window to find dispensed materials and print labels for them. You can reverse dispense from this window. Following is the procedure to search for dispensed material.

Prerequisites

- Dispense material.
- Optionally, identify the requirement to print labels for dispensed material.

To search for dispensed material:

1. Navigate to the Dispensed Material Activity window.
2. Enter as many parameters as needed to complete the search:

- Select an Organization from the list. Required.
 - Search and select a Dispense Area from the list by Dispense Area name or Description.
 - Search and select a Batch from the list. The list displays only the pending and work in process (WIP) batches in the selected organization.
 - Select a Batch Step number.
 - Search and select an Ingredient by Name or Description.
 - Search and select a Product from the list by Name or Description.
 - Narrow your search by selecting the material dispense date as, From Date, To Date.
 - Narrow your search using the Dispense Number. This is the dispense document number. It is valid only if you have configured manual document number entry.
3. Select the reservation Status. The available options include All, Deleted Reservations, and Reverse Dispense Only.
 4. Select Show Zero Net Dispensed Quantity Rows to display all items not yet dispensed.

Additional Information: Deselect this option for reverse dispensing operations.
 5. Click Go.

Viewing the Dispensed Material Activity Details

Run any pending instructions, print labels, and reverse dispense from the Dispensed Material Activity Details window. Following is the procedure to view the dispensed material activity details. Use this window to:

- Determine if all instructions were performed.
- View the Reverse Dispense History.
- Associate scanned labels to their corresponding label history record. This record becomes part of the Control Batch Record (CBR) when you select the GMO Label category before you upload the scanned label images.

Prerequisites

- Create dispensable material.
- Optionally, run pending instructions, print labels, or scan labels for upload to the dispensed material record.

To view the dispensed material activity details:

1. Search for dispensed material from the Dispensed Material Activity window.
Refer to "Searching for Dispensed Material" for detailed information.
2. Click Details. The Dispensed Material Activity Details window displays for the selected material:
 - Organization is the abbreviated name of the context organization.
 - Organization Name is a brief description of the organization.
 - Dispense Area is the code that identifies the subinventory where the batch consumes the dispensed material.
 - Ingredient Description is a brief description of the material dispensed.
 - Dispense UOM is the unit of measure for the dispensing equipment.
 - Lot is the code used to identify the
 - Hazardous Material (Yes or No)
 - Hazard Class is the classification for a hazardous material.
 - Material Status is Dispensed or Reverse Dispensed.
 - Batch is the batch document number.
 - Batch Step is the batch step identified with this dispense operation.
 - Operation is the code used to identify the dispense operation.
 - Line Num is the batch sequence line number for the operation.
 - Product is the code used to identify the product.
 - Description is a brief description of the product.
 - Source Device is code assigned to the source dispensing device.

- Description is a brief description of the source device.
- Dispense
 - Dispense Number is the document number assigned to the dispense operation.
 - Mode is the dispense mode used. Options are Full Container, Source Container, or Target Container.
 - Source Device or Target Device is the name of the device used for the dispensing operation.
 - Description is a brief description of the source or target device.
 - Requested Quantity is the total quantity needed for the dispensing operation in the selected dispense mode.
 - Material Loss is the quantity of material lost due to a problem other than the actual dispensing operations, such as spillage or equipment failure.
 - Reverse Dispensed Quantity is the quantity of material removed from the container holding the dispensed material for adjustment purposes.
 - Net Dispensed Quantity is the actual quantity of material dispensed in the selected mode.
 - E-record is the e-record associated to the dispensing operation.
- Containers
 - Source Container or Target Container is the item code associated to the container.
 - Description is a description of the container.
 - Source Container Tare or Target Container Tare is the tare weight of the source container or target container.
- Measures
 - Measure of Source Container is the number of times the source container was measured.
 - Quantity Dispensed is total quantity dispensed from the source container.
 - Measure of Target Container is total quantity dispensed from the source

container.

- Quantity Dispensed is total quantity dispensed from the source container.
- Labels
 - Security Tag is the alphanumeric tag used to label the dispensed material.
 - Dispense Label indicator is used to select the dispense label for printing.
 - Material Label indicator is used to select the material label for printing.
 - Pallet Label indicator is used to select the pallet label for printing.

Additional Information: Select the labels to print and click Print Labels.

- Dispense Instructions
 - Instruction Set is the name of the instruction set for the dispensed material activity.
 - Description is a brief description of the instruction set.
 - SOP is the standard operating procedure associated to this dispensed material activity.
- Instruction
 - Click Details to display complete instruction information.
 - Status is the instruction status.
 - Instruction Num is the instruction number. This number is assigned when you set up the application numbering.
 - Instruction Type is the instruction type.
 - Instruction Text is the instruction text entered.
- Label Printing History
 - Label Type is the type of label printed. Seeded label types include: Material Label for Raw Materials, Product Label Sample Label, Dispensed Ingredient Label, and Pallet Label.

- Print Count is the total number of labels printed for the label type associated to this dispensing operation.
- Print Date is date this label type was printed.
- Printed By is the name of the resource that printed the labels.
- Status is the print status for this label type.
- E-record is the e-record associated to printing this label type.
- Attachments are the graphical representations of this label.
- Reverse Dispense History
 - Item is the material dispensed as it appears on the inventory Item Master.
 - Description is a brief description of the material.
 - Reverse Dispense Number is the reverse dispense document number. It is valid only if you have configured manual document number entry.
 - Lot is the item lot associated to this reverse dispense action.
 - Batch is batch associated to this reverse dispense action.
 - Batch Step is the batch step associated to this reverse dispense action.
 - Operation is the operation in the listed batch step that is associated to the performance of this reverse dispense action.
 - Quantity is the quantity of material reverse dispensed.
 - UOM is unit of measure of the quantity.
 - Reverse Dispensed Date is date and time stamp of the reverse dispense action.
 - Click Details for detailed information on this reverse dispense number.

Viewing Reverse Dispense Instructions

Following is the procedure to view reverse dispensing instructions.

Prerequisites

- Enter reverse dispense instructions.

- Dispense material.

To view reverse dispense instructions for the dispensed material:

1. Search for a dispensed material from the Dispensed Material Activity window.
Refer to "Searching for Dispensed Material" for detailed information.
2. Select an item for reverse dispensing.
3. Click Reverse Dispense.
4. Perform tasks as defined in the instructions.
5. Click Finish.

Performing Reverse Dispensing

There are several reasons to perform reverse dispensing in production operations. Some typical reasons are:

- A parent batch is cancelled or scaled down after material is dispensed.
- To perform quality testing on material subjected to adverse conditions during transit to the shop floor.

Following is the reverse dispensing procedure.

Prerequisites

- Dispense material.
- Optionally, run pending instructions, print labels, or scan labels for upload to the dispensed material record.

To reverse dispense reserved material using the Full Container mode:

1. Navigate to the **Reverse Dispense Material** window
2. Enter the **Dispense Number**. This is the dispense document number. It is valid only if you have configured manual document number entry. Required.
3. Select **Booth** for the chosen Dispense Area.
4. Select dispensing **Mode** as **Full Container**.
5. Search for and select the **Source Scale**.

A box opens, displaying the available scales. Use the **Filter By** option to narrow the

list of scales by:

- Dispense Area
- Booth. Select Booth to view only the scales linked to the booth selected in the preceding step.

6. Search and select the Source Container by **Container Name** or **Description**.

Additional Information: Container information is set up in the Oracle Inventory Item Master.

7. Click **Read Measure** to measure the source container weight. The reverse dispense weight is calculated.
8. Enter a **Security Tag** number.
9. Select **Dispense Label** and **Material Label** to indicate label formats for printing.

Reverse Dispense Material Page Using the Full Container Mode

The screenshot shows the 'Reverse Dispense Material' page for 'PVI2, PVI2 LOT(Pharmaceutical RM, PR1 PR1)'. The page is divided into several sections:

- Quantity:** Shows 'Quantity Dispensed 10 LB' and 'Quantity Reverse Dispensed 9 LB'.
- Selection:** Includes a 'Booth' dropdown set to 'PHARMA Dry Goods', a 'Mode' dropdown set to 'Full Container', and an 'Include Tare' toggle switch.
- Dispense Context:** Displays 'Batch 277', 'Lot L01', 'Hazard No', and 'Hazard Class'.
- Source:** Contains a 'Scale' dropdown set to 'SCALE2', a 'Container' dropdown set to 'CONTAINER-SD2', a 'Tare' input field set to '1', and a 'Quantity' input field set to '10' with a unit dropdown set to 'LB'.
- Material Loss:** Includes a 'Quantity' input field and a unit dropdown set to 'LB'.
- Security:** Includes a 'Tag' input field.
- Label:** A table with columns 'Select', 'Label Type', 'Printer', and 'Label Count'. It lists 'Dispense' and 'Material' label types, each with a 'Select' checkbox and a 'Label Count' input field set to '1'.

At the bottom right, there are 'Cancel' and 'Continue' buttons. The footer includes copyright information and links to 'About this Page' and 'Privacy Statement'.

10. View the reverse dispense instructions.
11. Obtain e-signatures.
12. Print the labels.

To reverse dispense reserved material using the Target Container mode:

1. Navigate to the **Reverse Dispense Material** window.
2. Enter the **Dispense Number**. It is valid only if you have configured manual document number entry. Required.
3. Select **Booth** for the chosen Dispense Area.
4. Select dispensing **Mode** as **Target Container**.
5. Search for and select the **Source Scale**.
A box opens, displaying the available scales. Use the **Filter By** option to narrow the list of scales by:
 - Dispense Area
 - Booth. Select Booth to view only the scales linked to the booth selected in the preceding step.
6. Click **Read Measure** to capture readings directly from an integrated device.
7. Search and select the **Target Container** by **Container Name** or **Description**.
8. Measure the target container weight with the material transferred. The reverse dispensed quantity is calculated and displayed.
9. **Re-measure Source Container** weight to verify the reverse dispensed quantity.

Additional Information: If the source container weighs less than expected due to unavoidable loss of material, you must manually enter the quantity discrepancy in **Material Loss**.
10. Enter a **Security Tag** number.
11. Select **Dispense Label** and **Material Label** to print labels.

Reverse Dispense Material Page Using the Target Container Mode

The screenshot shows the 'Reverse Dispense Material' page for material PVI1, PVI1 (Pharmaceutical RM, PR1 PR1). The page is titled 'Reverse Dispense Material Page Using the Target Container Mode'.

Quantity: Quantity Dispensed: 9 LB, Quantity Reverse Dispensed: 5 LB.

Selection: Booth: PHARMA Dry Goods, Mode: Target Container, Include Tare: ☒.

Dispense Context: Batch: 291, Lot: , Hazard: No, Hazard Class: .

Source: Scale: SCALE2, Container: CONTAINER-SD2, Tare: 1, Quantity: 9 LB.

Target: Scale: SCALE2, Container: CONTAINER-SD2, Tare: 1, Quantity: 6 LB.

Reweight Source: Quantity: 4 LB.

Material Loss: Quantity: , LB.

Security: Tag: .

Label: A table with columns: Select, Label Type, Printer, Label Count.

Select	Label Type	Printer	Label Count
☐	Dispense		1
☐	Material		1

Buttons: Cancel, Continue.

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12. View the dispense instructions.
13. View the Partial Dispense History report.
14. Obtain e-signatures.
15. Print the labels.

Viewing the Reverse Dispensed Material Activity Details

Run any pending instructions, or reprint the reverse dispense labels. Following is the procedure to view the reverse dispensed material activity details.

Prerequisites:

- Reverse dispense material.

To view the reverse dispensed material activity details:

1. Navigate to the Dispensed Material Activity Details window.
2. View the Reverse Dispense History of an item.
3. Click Details to display the material activity details.

Note: Refer to "Viewing the Dispensed Material Activity Details" for explanations of the fields displayed on the Reverse Dispensed Material Activity Details window.

Running Reports for Dispensing Operations

The Dispense Dispatch report lists all the pending dispenses. The Dispense History report lists all the completed dispense operations.

To run and print the Dispense Dispatch report:

1. Navigate to the Submit a New Request window:
Select MES for Process Manufacturing Administrator: Request : Run.
2. Select the request type as Single Request.
3. Select OK.
4. Enter Dispense Dispatch Report as the request Name.
If you are submitting a request set, then enter the Request Set name in the Submit Request Set window.
5. Complete the other fields on the window. See: Overview of Standard Request Submission, *Oracle E-Business Suite User's Guide*
6. Select the various parameters such as Plant, Dispense Area, Batch, From and To dates.
7. Select Submit Request.

Following is the output of the Dispense Dispatch report:

- Organization
- Search Criteria
 - Batch is the batch document number.
 - Plant is the name of the production facility where the material is dispensed.
 - Dispense Area is the inventory source for the dispensed material.
 - From Date is the start date for this report.
 - To Date is the end date for this report.
- Dispensable Material
 - Item is the material dispensed as it appears on the inventory Item Master.
 - Description is a brief description of the material.

- Lot is the item lot associated to this dispense action.
- Batch No is the batch document number.
- Batch Step is the batch step associated to this dispense action.
- Operation is the operation in the listed batch step that is associated to the performance of this dispense action.
- Required Quantity is the prescribed quantity for the dispense operation.
- UOM is the unit of measure for the dispensing equipment.
- Required Date is the date and time the dispense operation is required.

To run and print the Dispense History report:

1. Navigate to the Submit a New Request window:
Select MES for Process Manufacturing Administrator: Request : Run.
2. Select the request type as Single Request.
3. Select OK.
4. Enter Dispense History Report as the request Name.
5. Complete the other fields on the window. See: Overview of Standard Request Submission, *Oracle E-Business Suite User's Guide*.
6. Select the appropriate parameters for the report such as Plant, Dispense Area, Batch, From and To dates.
7. Select Submit Request.
To submit another request, select Yes.
8. Following is the output of the Dispense History report:
 - Organization is the abbreviated name of the context organization.
 - Dispense Area is the code that identifies the subinventory where the batch consumes the dispensed material.
 - Search Criteria
 - Batch is the batch document number.

- Plant is the name of the production facility where the material is dispensed.
- Dispense Area is the inventory source for the dispensed material.
- From Date is the start date for this report.
- To Date is the end date for this report.
- Operator is the name of the person who performed the dispensing operation.
- Dispensed Material
 - Item is the material dispensed as it appears on the inventory Item Master.
 - Description is a brief description of the material.
 - Dispense Number is the dispense document number. It is valid only if you have configured manual document number entry.
 - Lot is the item lot associated to this dispense action.
 - Batch is the batch document number.
 - Batch Step is the batch step associated to this dispense action.
 - Operation is the operation in the listed batch step that is associated to the performance of this dispense action.
 - Line Number is the batch sequence line number for the operation.
 - Net Dispensed Quantity is the actual quantity of material dispensed in the selected mode.
 - Dispensed Date is the date and time stamp of the dispense action.
- Required Quantity is the prescribed quantity for the dispense operation.
- Dispense Mode is the dispense mode used. Options are Full Container, Source Container, or Target Container.
- Dispense UOM is the unit of measure for the dispensing equipment.
- E-record ID is the e-record associated to this dispense operation.
- Quantity Reverse Dispensed is the quantity of material reverse dispensed using the dispense mode indicated.

- Material Loss is the quantity of material lost due to a problem other than the actual dispensing operations, such as spillage or equipment failure.
- Operator is the name of the person who performed the dispensing operation.

Device Integration

This chapter covers the following topics:

- Understanding Device Integration
- Mapping a Device to a Resource
- Using Integrated Devices to Enter Process Parameters

Understanding Device Integration

Oracle MES for Process Manufacturing uses the device integration capability of Oracle Warehouse Management System to read quality test results, process parameters, and weighing scales. Full mobile device support provides a query for dispensable items and quality test results directly from a handheld device, and selection of dispensing jobs. You can read data from the device with a single click.

Most data entry required for dispensing and preweigh is related to specifying material details such as item number, and lot and quantity information. Barcode readers to scan item and lot information provide data entry capability and eliminate data entry errors. Standard weighing scale integration is provided to read the quantities data directly from production scales or balances.

- You can sign onto one or multiple devices prior to accepting tasks. When you sign on to a device you unlock it for use in the manufacturing process. Your system administrator configures the device to require sign-on prior to task dispatch.
- You can flag devices that require exportation of lot-specific information, and include this information in the data transmission.

Setup procedures are documented in the mobile device user's guide for configuration information.

Related Topics

Tasks of a Database Administrator, *Oracle Database Administrator's Guide*

Configuring Dispatchers, *Oracle Database Administrator's Guide*

Defining Devices, *Oracle Warehouse Management User's Guide*

Explaining Material Handling Devices, *Oracle Warehouse Management User's Guide*

Material Handling Device Integration, *Oracle Warehouse Management Implementation Guide*

Entering Quality Results Using the Touch Screen Operator Workbench, page 13-9

Mapping a Device to a Resource

The Oracle Warehouse Management material handling device integration lets you manage data through several different methods. After mapping a dispenser to a resource, for example, you can export a measurement command to a dispenser and import measurement data from for processing to MES for Process Manufacturing.

Prerequisites:

- Define material handling devices. See: Defining Devices, *Oracle Warehouse Management User's Guide* and Explaining Material Handling Devices, *Oracle Warehouse Management User's Guide*.
 - Associate devices at the subinventory level.
 - Associate the message template with the device definition.
 - Specify the PL/SQL API specification interface for the device. The export method uses a PL/SQL package procedure called from Warehouse Management in the context of processing the business event.
- Link the material handling device to one or more business events. See: Material Handling Devices and Business Events, *Oracle Warehouse Management User's Guide*.

Note: You can map a single device for both the Process Parameter Event and the Process Dispensing Event.

- The Process Parameters Event is for mapping the device to acquire process parameters such as temperature, pressure, humidity, and other critical environmental variables.
- The Process Dispensing Event is for mapping the device to a dispenser.
- The Process Quality Result is for mapping the device to a quality test.

To map a device to a dispensing or process parameter resource:

1. In the **Process Engineer** responsibility, navigate to the **Plant Resources** window.
2. Enter the organization.
3. Query the desired resource.
4. Select devices. The **Resource Devices** window appears.
5. Enter device information.
6. Enter the device response to the process parameter to map the device to the resource.
7. Save your work.
8. Verify the device mapping by performing an appropriate task that requires entry of a process parameter or performance of a dispensing-related operation.

To map a device to a lab test resource:

1. From the Process Engineer responsibility, select **Setup**, and then **Plant Resources**.
2. Select a laboratory organization.

Important: Before mapping a device to a lab test resource, verify that the resource is assigned to a laboratory organization. See: *Defining Plant Resource Parameters, Oracle Process Manufacturing Process Planning User's Guide*.

3. Query the lab test resource.
4. Click **Instances**. Verify that the resource instance you want to map to the device exists. If it does not exist, create the instance.
5. Navigate to the **Test Methods** window. Select a test method, then click **Resource**.
See: *Entering Test Methods, Oracle Process Manufacturing Quality Management User's Guide*.
6. On the **Resources** window, add one or more resource, resource instance, and laboratory organization combinations to the test method.
7. Select a default resource instance for each laboratory organization using that test method.

8. Navigate to the **Resource Device Mapping** window using the Quality manager responsibility.
9. Map the device response fields to the test.
See: Using the Resource Device Mapping Window, *Oracle Process Manufacturing Quality Management User's Guide*.
10. Click Save.
11. Verify the device mapping by reading the device while entering quality results for a batch. See: Entering Quality Results Using the Touch Screen Operator Workbench, page 13-9.

Related Topics

Using Integrated Devices, page 6-4

Setting Up and Integrating Device Reading, *Oracle Process Manufacturing Quality Management User's Guide*

Using Integrated Devices to Enter Process Parameters

A process parameter is entered to record information such as temperature, humidity, relative barometric pressure, or elapsed time pertinent to product manufacturing. Use process parameters as set points, comparison values, or to document environmental conditions or the measured duration of an operation. See: Mapping a Device to a Resource, page 6-2

To measure process parameters using an integrated device:

1. Navigate to the Production Supervisor responsibility.
2. Create a batch from a recipe that requires entry of process parameters.
3. Navigate to the Batches window to release the batch.
4. Navigate to the Batch Steps window to release the batch step that has the requirement for process parameter entry.
5. Select Activities to display the Step Activities window.
6. Select the desired activity.
7. Select Resources to display the Step Resources window.
8. Select the desired resource.

9. Select Process Parameters to display the Resource Process Parameters window.
10. Select the desired process parameter.
11. Select Read Selected Process Parameter to trigger the measurement and data entry sequence.

The application reads the parameter from the integrated device and communicates the reading to the application. The Actual process parameter displays.

Nonconformance Tracking

This chapter covers the following topics:

- Requirement
- Understanding Nonconformance Tracking
- Capturing Material Nonconformances
- Capturing Resource Nonconformances
- Disposing Nonconformances
- Logging Nonconformances

Requirement

Following is a requirement that customers have for tracking nonconformances:

Does the application support nonconformance tracking?

Yes. Upon identifying a nonconformance, suppliers should document, evaluate, segregate (where practical) and disposition the nonconformance in accordance with the defined nonconforming material control process. Oracle Manufacturing Execution System (MES) for Process Manufacturing provides the toolset for rapid identification and tracking of manufacturing nonconformances.

Capture and Process Material and Resource Nonconformances

Written procedures for production and process control are designed to assure that critical materials such as drug products have the identity, strength, quality, and purity they claim to have or are represented to possess. Oracle Manufacturing Execution System (MES) for Process Manufacturing helps ensure that written nonconformance procedures, including any changes, are drafted, reviewed, and approved by the appropriate organizational units and reviewed and approved by the quality control unit. Electronic records are maintained with electronic signatures so that these process control procedures are followed in the execution of the various production and process

control functions. Tasks are documented at the time of their performance. Any deviation from the written procedures is then captured, recorded, and an appropriate disposition is assigned.

Understanding Nonconformance Tracking

Unexpected events that adversely affect the manufacturing process and quality of the finished product can occur in any controlled production environment. These nonconformances must be tracked in the Control Batch Record (CBR). Quality departments analyze nonconformances as part of their Corrective Action and Preventive Action initiatives (CAPA). To document the complete manufacturing batch record, evidence of nonconformances must be included in the CBR. Oracle Manufacturing Execution System (MES) for Process Manufacturing helps you manage nonconformance in the batch manufacturing process.

Nonconformances observed during the following events are tracked and recorded in the CBR and are used for analysis by Quality Control for CAPA. When there is nonconformance from the Master Batch Record (MBR), a window displays the following details:

- Time the event occurred
- Batch number, product produced
- Phase of the batch where the event occurred
- Specific details of the incident or observation, including:
 - Date of the nonconformance
 - Level of criticality
 - Individual who logged the issue
 - Type of nonconformance
 - Target value (not required)
 - Actual value
 - Preliminary cause
 - Reason code or severity code
 - Remarks
 - Links to standard operating procedures for managing nonconformance

- Required follow-up steps with signatures of responsible operators and supervisors

Classify nonconformances as either critical or noncritical. Specify the actions to take after a nonconformance is recorded. For example, critical nonconformance is signed off by a quality control manager, while the production supervisor signs off noncritical nonconformance.

If additional follow-up and analysis documentation is created, then it is associated with the batch and incorporated into the CBR. Display the SOPs that link to a specific type of nonconformance.

Each nonconformance is identified by a nonconformance number and is available for viewing from the CBR.

Log a dynamic nonconformances during the batch execution process by assigning nonconformance severity, priority, and nonconformance ownership.

Oracle Manufacturing Execution System (MES) for Process Manufacturing tracks manufacturing nonconformances caused by such things as batch material quantity changes, item substitutions, resource substitutions, as well as any ad-hoc operator observations, such as power interruptions. Notifications of any nonconformances are automatically routed electronically to those who need to review and approve or disapprove continued processing. This includes all appropriate nonconformance details and electronic signature information so that batch status can be determined and appropriate actions can be taken. Batches with unapproved critical nonconformance events are now allowed to proceed to a closed status until all critical nonconformance events have received an approval disposition.

Capturing Material Nonconformances

If batch material quantities are changed or substituted in the manufacturing process, then a nonconformance is recorded manually in the Oracle Process Manufacturing Process Execution application by choosing Log Nonconformance from the Actions menu. This can occur when the dispense material workorder is modified to change dispense quantities prior to staging material for addition on the shop floor. It can also occur when batch scaling is required to correct for material discrepancies.

This is a critical nonconformance. It must be authorized by the quality manager. Notifications of this nonconformance are sent to individuals who need to review and approve or disapprove the continuance of batch processing.

Capturing Resource Nonconformances

If resources are changed during batch production, then a nonconformance is recorded. By definition, a resource is anything of value, except material and cash, required to manufacture, cost, and schedule products. Resources include people, tools, machines, labor purchased from a supplier, and physical space used in the manufacturing process.

Resource nonconformances are defined as:

- Resource substitutions in the batch manufacturing process that were not previously defined as acceptable alternates.
- Resource failures and performance issues

This is a critical nonconformance and must be authorized by the quality manager.

Disposing Nonconformances

A workflow approval process is available to make disposition on nonconformance. The post disposition actions such as putting a batch on hold, changing grade, and so on are manual.

Refer to the *Oracle Quality User's Guide* for details.

Logging Nonconformances

A collection plan and rules are seeded for Oracle Manufacturing Execution System (MES) for Process Manufacturing. Nonconformance e-records and associated e-signatures are part of the Oracle E-Records store for each event tracked. You can log nonconformances at the following batch levels:

- Batch
- Batch operation step
- Batch operation step activity
- batch operation step activity resource

Navigate to the following window or pages to log a nonconformance from the:

- Batch Steps window in the Production Supervisor responsibility of the OPM Process execution application.
- Batch page, Batch Details tab of the Operator Workbench.
- Batch Operation page of the Operator Workbench:
 - Batch Details tab
 - Batch Details tab, Step
 - Batch Details tab, Step, Activity
 - Batch Details tab, Step, Activity, Resource
- Process Instructions page of the Touchscreen Workbench.

From here, you can log a nonconformance from the batch, step/operation, activity, or resource level.

Prerequisites:

- Set up nonconformance tracking in the Oracle Quality application.
Refer to Setting Up Nonconformance Tracking, page 2-15 for additional information.
- Identify the batch, operation, step, activity, or resource where the nonconformance occurred.

To log a nonconformance from the Batch Steps window:

1. Navigate to the Batch Steps window.
2. Select the Step and Operation where the nonconformance was observed.
3. Choose Log Nonconformance from the Tools menu. The Enter Quality Results window displays.
4. Verify the Collection Plan, Owner and Enter By User fields. You can only select a collection plan that has a current effective date. If a default collection plan has been specified, it is automatically selected.
5. Click Enter. The Update Quality Results window displays.
6. Enter the Nonconformance Number.
7. Enter Nonconformance Source as INPROCESS. Required.
8. Enter a brief Description of the nonconformance. Required.
9. Enter a Detailed Description of the nonconformance. Required.
10. Click Update.

To log a batch nonconformance from the Operator Workbench:

1. Navigate to the Operator Workbench.
2. In the Operator Workbench, find and select a batch or batch operation.
3. In the Batch or Batch Operation page, select the Batch Details tab.
4. In the Actions field, select Log Nonconformance.

5. Click Go.

The Quality Plans page displays.

Continue entering the batch nonconformance as described in the following task *To enter a nonconformance*;, page 7-7.

To log a batch step nonconformance from the Operator Workbench:

1. In the Operator Workbench, find and select a batch operation.
2. In the Batch or Batch Operation page, select the Batch Details tab.
3. In the Operations region, select a step.
4. In the Select Step field, select Log Nonconformance.
5. Click Go.

Continue entering the batch step nonconformance as described in the following task *To enter a nonconformance*;, page 7-7.

To log a batch step activity nonconformance from the Operator Workbench:

1. After selecting a batch step as described in the preceding task *To log a batch step nonconformance from the Operator Workbench*;, in the Activities region, select an activity.
2. In the Select Activity field, select Log Nonconformance.
3. Click Go.

Continue entering the batch step activity nonconformance as described in the following task *To enter a nonconformance*;, page 7-7.

To log a batch step activity resource nonconformance from the Operator Workbench:

1. After selecting a batch step activity as described in the preceding task *To log a batch step activity nonconformance from the Operator Workbench*;, in the Resources region, select a resource.
2. In the Select Resource field, select Log Nonconformance.
3. Click Go.

Continue entering the batch step activity resource nonconformance as described in the following task *To enter a nonconformance*;, page 7-7.

To log a nonconformance from the Touchscreen Workbench:

1. Navigate to the Touchscreen Workbench.
2. Search for a batch.
Refer to Searching for Batches, page 13-3.
3. Touch, tap or press Details to view process instructions for a batch or batch operation step.
4. If you are viewing process instructions for a batch in the Batch page, touch, tap or press Log Nonconformance.
If you are viewing process instructions for a batch operation step in the Batch Operation page, select the level to enter the nonconformance:
 - Batch/Material
 - Step
 - Activity
 - ResourceTouch, tap or press Log Nonconformance.
5. Continue entering the batch step activity nonconformance as described in the following task To enter a nonconformance:, page 7-7.

To enter a nonconformance:

1. In the Quality Plans page, click Enter Quality Results for a collection plan.
The Add Data: Enter Results page displays.
2. Verify the Plan Name and Process Batch Number.
3. In the Nonconform Severity field, select the severity level.
4. (Optional) In the Detailed Description field, describe the nonconformance.
5. (Optional) Click Attachments to add an attachment.
6. (Optional) Click Add Another Row to add another nonconformance.
7. Click Apply.

Label Printing

This chapter covers the following topics:

- Requirements
- Understanding Label Management
- Printing Dispense Labels
- Printing Labels from a Touchscreen

Requirements

Process manufacturers need to print and track labels for ingredients, samples, and products.

Is label printing and tracking supported?

Yes. Labels mark the contents of containers in organizations that produce controlled materials such as pharmaceuticals and fine chemicals. Oracle Manufacturing Execution System (MES) for Process Manufacturing prints and tracks labels for raw materials products, intermediates, and quality samples that are part of the production process. Track label reprinting with a reprint number that appears on the actual label or reprint a label without tracking if the original printed label is unusable.

Produce Accurate Labels for Every Part of the Production Process

Oracle Manufacturing Execution System (MES) for Process Manufacturing prints labels, and determines whether a label was previously printed for a defined set of key values such as a raw or intermediate material, batch number, product number, or sample number. If the label was previously printed, then a reprint number is created to track the label printing operation. The reprint number is printed on the label itself to indicate that more than one copy exists. However, if you know that the original label is unusable, you also have the option to reprint the label without a reprint number. A variety of data elements are available for customized label design.

Understanding Label Management

The flow of raw materials, components, and finished goods must be tracked and managed throughout the supply chain. Tracking raw materials, ingredients, intermediates and finished goods throughout production requires timely information capture and transfer to improve production speed and efficiency. Production tracking to the unit or material level helps improve efficiencies at every level within the production process. Barcode printers and readers are used in the context of many business events including dispensing, in-process quality sampling, and batch output. Oracle Manufacturing Execution System (MES) for Process Manufacturing provides label support that integrates third-party label printing software packages with components from the Oracle Warehouse Management System (WMS). The application extracts data from tables, and manages it with Extensible Markup Language (XML) formats to print custom labels. Configure individual application business events to enable user-initiated or automatic label printing. Label count and reprint options are available. XML provides fully configurable label output. Embedded barcode images in the Control Batch Record assist operators on the plant floor to retrieve batch information quickly and reliably using mobile barcode readers.

Oracle Manufacturing Execution System (MES) for Process Manufacturing uses the label-printing solution available in Oracle Warehouse Management System (WMS) to print customized labels.

Refer to "Compliance Labeling Setup" and related topics in the *Oracle Warehouse Management Implementation Guide* for additional information.

Labels available include:

- Material Label for Raw Materials
- Product Label
- Sample Label
- Dispensed Ingredient Label
- Pallet Label

Major Features

Configure the application to print selected labels automatically. Copies of the labels affixed to containers are retained and archived in image or PDF formats. Print or reprint labels at various points in a business scenario. When you reprint, the labels display the print count to indicate the total number of times the label was printed, although you also have the option to reprint an unusable label without increasing the print count. You can choose to require authorized approvers to approve the label printing or reprinting. Configure the label content as described the *Oracle Warehouse Management*

Material Label

The material label is used primarily for identifying raw materials in the manufacturing process. The label is printed for raw materials moved to an inventory organization during purchase receipt. It is also printed to label materials dispensed into containers, or to identify materials that are reverse dispensed and returned to stores.

Data presented on a material label includes:

- Item Number and Description
- Lot
- Date Received
- Expiration Date
- Hazardous Material Classification

Triggering Events

Material label printing is triggered by the following business events:

- Raw Material Receipt

As part of the receiving process, labels are placed on the pallets before they are sent to the warehouse. Label printing is triggered by a variety of configurable business events, including the reading of a Radio Frequency Identification (RFID) tag, scanning a barcode, or manual entry of a standard receipt or a direct receipt against a purchase order.
- Full Container Dispense

Label printing is triggered by dispensing the full container.
- Source Container Dispense

Excess material is removed from the source container and placed in an empty target container. The source container is used in the production batch. The target container is returned to inventory and so must be labeled with this label. Label printing is triggered by saving the source container dispense record.
- Target Container Dispense

Material is removed from the source container and placed in a target container. The target container is used in the production batch. The source container is returned to inventory, if it has material remaining in it. This container must be relabeled to indicate the new quantity. Label printing is triggered by saving the target container dispense record.

You can also manually print material labels from the Touch Screen Print Label page.

Product Label

Print product labels manually for lots reserved to a batch. This is useful when you preallocate product lots for a batch and print the labels in advance. The product label is generated for all products produced as part of the batch. Data elements in a product label are:

- Product Number
- Product Description
- Plant Code
- Batch Number
- Lot
- Quantity Produced
- Date of Manufacture as the Lot Creation Date
- Expiration Date
- Hazardous Material Class
- Notes

Sample Label

Print sample labels automatically while performing quality tasks or print them manually using the Touchscreen Print Label navigation. The sample label is generated when the intermediates or products are yielded and marked for clearance by quality control. The data elements that are present in a product label are:

- Intermediate or Product Number
- Intermediate or Product Description
- Sample Number
- Lot
- Quantity
- Date Sampled

- Date of Manufacture, that is, Lot Creation Date
- Expiration Date
- Hazardous Material Class
- Notes

Triggering Event

Quality sample labels are printed automatically when a sample record is saved.

Pallet Label

Dispensed raw materials for each individual batch are stored in individual cages. These cages are identified using the Batch Collation Label. This is generated when the first dispense of the material in a batch occurs.

Label Storage

All labels of ingredients, intermediates, products, and samples are stored against each batch record. In addition to storing the labels generated by the application, you can attach labels scanned from hard copies produced for a specific batch.

Security Code on Labels

A security code is used for controlling raw material for drugs. When controlled drugs are dispensed, the bags are sealed with a security tag. This number is recorded against the Dispense field in the Dispensing window that prints out on the label. The same number is also recorded in batch records, enabling a three-way match of the security code for the seal code, label, and batch record. This ensures that the container seal is not tampered.

Handling Code

The handling code is the hazard code attached to a raw material that indicates the precautions to take when dispensing or handling it.

Printing Dispense Labels

To print dispense labels:

1. Navigate to Dispensing.
2. Select the Label Type to print and dispense the material.

Additional Information: Selecting the label type is not necessary if autoprint configuration is enabled.

3. Select Apply and obtain the e-signature.

Scan and attach the label to the dispense record from the Dispensed Material Activity Details window, if you want to retain the copy of the label in Oracle Applications. The Label Details window displays the history of labels printed earlier for a business flow or for a given label type.

Printing Labels from a Touchscreen

Use the Touchscreen Print Labels page to manually print the following labels without a reprint number if, for example, a previously printed label is damaged.

- Dispense label
- Pallet label
- Product label
- Sample label

To manually print labels from the touch screen user interface:

1. Navigate to the Print Label page.
2. Select the appropriate organization from the Organization list.
3. Select one of the following choices from the Business Flow list:
 - Process Dispensing: searches for completed dispense transactions.
 - Process Production: searches for product transactions.
 - Process Quality: searches for samples drawn.

4. Touch, tap or press Accept to open the Print Labels page.

The Print Labels page displays all transactions that meet the default criteria for the business flow selected. If you do not need to change the search criteria, go to step 8.

To change the business flow search criteria

5. In the Search Criteria region, touch, tap or press Change to change the default search criteria for the selected business flow transactions.

The Select Search Criteria page opens. The search criteria fields differ based on the business flow selected.

6. Enter your search criteria in the appropriate fields.
7. Touch, tap, or press Accept to return to the Print Labels page and view the search results.

To print a label for a transaction

8. Expand the details of a transaction.

The Label Type displays.

Conditional Action: For the Process Dispensing business flow, if there are multiple ingredients to dispense, select whether to print a Dispense (for one ingredient) or Pallet (for all ingredients) label type.

9. In the Printer field, select the printer to print the label.
10. In the Label Count field, enter the number of labels to print.
11. Touch, tap, or press Print Label.

If no e-signatures are required, this completes the task.

To approve label printing

12. In the E-signatures page, enter your User ID and Password.
13. Optionally, change the default values for Signature Type and Signing Reason.
14. Optionally, enter Comments.

Optional Action: Select the E-Record tab to view the Label Print Request details.

15. Touch, tap, or press Approve.
16. Touch, tap, or press Apply.

A Confirmation appears and provides the label print request ID.

To view the label print request:

From the Warehouse Manager responsibility, select Inquiry and then View Label Requests.

Use the View Label Requests window to find your label print request ID, view the label status, and so on.

Batch Records

This chapter covers the following topics:

- Requirement
- Understanding Batch Records
- Viewing and Printing the Master Batch Record
- Viewing the Production Status
- Viewing the Batch Progression
- Viewing and Printing the Control Batch Record

Requirement

Following is a requirement that customers have for supporting electronic batch records:

Does the application support electronic batch records?

Paper batch records are replaced by an electronic procedure management and recordkeeping system. The application provides a complete solution with electronic signatures and audit trails. The Electronic Batch Record (EBR) that defined by the US FDA, includes text, graphics, data, audio, pictorial or other information represented in digital form that is created, modified, maintained, archived, retrieved or distributed by a computer system. The Oracle Manufacturing Execution System (MES) for Process Manufacturing application lets you store all batch-relevant data in the database for rapid retrieval and processing.

Use the Master Batch Record and Control Batch Record for Paperless Recordkeeping

The master production record is often referred to as the master batch record. This is the documented and approved set of instructions used to describe how to manufacture a specific product. The required elements of the master batch record are outlined in CFR Title 21 Part 211.186.

The OPM master batch record consists of the recipe, which includes the material

requirements, equipment, resource requirements, clearance and processing instructions, sampling requirements and various notes regarding batch processing. It is the template that is used repeatedly to manufacture a consistent product.

The OPM control batch record is a journal that documents all events in the batch lifecycle. The control batch record aligns to the specifications of CFR Title 21 Part 211.188.

Understanding Batch Records

The U.S. Food and Drug Administration (FDA) provides guidelines for master production and control recordkeeping that meet current good manufacturing practice for finished pharmaceuticals:

- Master Batch Record (MBR) Section 211.186, Code of Federal Regulations (CFR) of the US Food and Drug Administration (FDA)
- Control Batch Record Section 211.188, Code of Federal Regulations (CFR) of the US Food and Drug Administration (FDA)

The intent of these records is to assure product uniformity from batch to batch. The master production and control records for each product include written procedures for batch formulation that must be followed on the shop floor, and a complete batch control record to document that each significant step in the manufacture, processing, packing, or holding of the batch is done. These records must be signed and dated by at least one individual, and independently checked, dated, and signed by a second person. Oracle Manufacturing Execution System (MES) for Process Manufacturing uses Oracle E-Records to provide a comprehensive, paperless method to comply with Title 21 Code of Federal Regulations (21 CFR Part 11) Electronic Records; Electronic Signatures.

Master Batch Record

The Master Batch Record (MBR) provides detailed recipe-specific information. It is a complete specification for manufacturing a specific batch size in a specific plant at a specific time. The MBR must provide:

- Name and strength of the product with its description
- Name and weight or measure of each active ingredient
- Complete list of components designated by names or codes sufficiently specific to indicate any special quality characteristics
- Accurate statements of the weight or measure of each component, using the same weight system
- Statement concerning any calculated component excesses

- Statement of theoretical weight or measure at appropriate phases of production processing
- Statement of theoretical yield, including the maximum and minimum percentages of theoretical yield beyond which an investigation is required
- Description of product containers, closures, and packaging materials, including a specimen or copy of each label and all other labeling signed and dated by the individuals responsible for label approval
- Complete manufacturing and control instructions, sampling and testing procedures, specifications, special notations, and precautions to be followed

The MBR contains all the key information in the product recipe, including a complete specification to manufacture the product in a given plant, batch size, and date range. The MBR contains the recipe header information, validity rule data, formula details, routing details, instructions, and material dispensing setup information with the corresponding instructions. It is available for viewing and printing after the recipe validity rule is approved for general use.

Control Batch Record

Batch control records provide a complete information set related to the production and control of each individual batch. The Control Batch Record (CBR) is updated with actual data as the batch proceeds through its lifecycle. Material usage, dispensing activities, and component weights and measure records provide batch traceability. Product output quantities, scrapped material, lot details, manufacturing nonconformances, and quality sampling and test results all add to this batch history. The CBR must provide:

- An accurate reproduction of the appropriate MBR checked for accuracy, dated, and signed
- Documentation that each significant step in the manufacture, processing, packing, or holding of the batch was accomplished, including:
 - Start date, stop date, and time stamps for each step
 - Identity of specific major equipment and production lines used
 - Identification of each batch of component or in-process material used
 - Weights and measures of components used in processing
 - In-process and laboratory control results
 - Inspection of the packaging and labeling area before and after use

- Statement of the actual product yield
- Statement of any scrapped ingredients, byproducts, or products
- Statement of the percentage of theoretical yield appropriate phases of batch processing
- Complete labeling control records, including specimens or copies of all labeling used
- Description of product containers and closure systems
- Quality sampling performed
- Identification of the individuals who performed each task
- Identification of individuals who supervised or checked each significant step in the operation
- Documentation of any investigations for nonconformance
- Results of examinations made for compliance

The electronic CBR is created when the production batch is created. It is updated with actual data as the batch progresses through its life cycle. The CBR contains information such as actual material usages, scrap created, product output quantities and lot details, manufacturing nonconformance, quality samples and results, label images, instructions, material dispensing setup information with the corresponding instructions, changes in batch and batch step status, operator comments, and all electronic signatures captured during the processing of the batch. Configure the system to obtain e-signature approvals before closing a batch.

The final stage of the production batch lifecycle is to close the batch. Oracle Manufacturing Execution System (MES) for Process Manufacturing documents this transaction with e-signature approvals. The final CBR is secured in the e-records repository, where authorized responsibilities can view or print it.

Title III of the US Public Health Security and Bioterrorism Preparedness and Response Act of 2002

Title III of the US Public Health Security and Bioterrorism Preparedness and Response Act of 2002 (also known as the "Bioterrorism Act") is "Protecting Safety and Security of Food and Drug Supplies." Title III requirements include "... the need to establish and maintain records by those who manufacture, process, pack, transport, distribute, receive, hold, or import food, as may be necessary to trace the source and the chain of distribution of food, its components and ingredients, and its packing in order to address threats of serious adverse health consequences or death to humans or animals." Oracle

Manufacturing Execution System (MES) for Process Manufacturing provides fully traceable recordkeeping that can be used to meet the requirements of this act. Coupled with other Oracle Process Manufacturing (OPM) applications such as Quality Management and Process Execution, build Hazard Analysis Critical Control Points (HACCPs) throughout the manufacturing supply chain. Oracle Manufacturing Execution System (MES) for Process Manufacturing helps meet the requirements for this important requirement for food and drug safety.

E-Record Events

The CBR captures the major events of the batch including dates of the event and name of the person who made the transaction for:

- New Material Added
- Material Removed
- Step Added
- Step Removed
- Additional Resource Requirements
- Deletion of Resource
- Activity Added
- Activity Update
- Resource Details Updated
- Batch Update (Header)
- Batch Step Update
- Process Parameter

Oracle Manufacturing Execution System (MES) for Process Manufacturing modifies the output of several events, including:

- Batch Release
- Material Allocation
- Step Release
- Step Reopen
- Batch Reopen

- Step Complete
- Step Close
- Batch Complete
- Batch Close
- Batch Termination
- Revert to Work in Process

Viewing and Printing the Master Batch Record

View and print the Master Batch Record (MBR) from Recipe and Recipe Validity Rules windows. The Master Batch Record (MBR) consists the recipe hierarchy:

- Recipe Summary Information such as Recipe and Recipe Validity Rule
- Recipe yield and yield % at step level
- Material Requirements such as Formula
 - Sample
 - Instructions with Standard Operating Procedure (SOP) Reference
 - Dispensing Setup
- Detailed Routing Information, that is, Routing Steps
 - Instructions with SOP Reference
 - Activities
 - Resource Requirements and Instructions with SOP Reference

Refer to "Recipes" in the *Oracle Process Manufacturing Product Development User's Guide* for detailed information on recipes and recipe validity rules.:

To view and print the MBR from a recipe:

1. Navigate to the Recipe Details window.
2. Create or query a recipe. Enter as many parameters as needed to complete the search:
Select Recipe, Description, Formula, Routing, or Product with appropriate Version

for Recipe, Formula, or Routing. Select Organization, Customer, From Date, To Date, Status, and Recipe Use.

3. Select View Master Batch Record from Actions.
4. Print the draft version of Master Batch Record.

To view and print the MBR from the recipe validity rule:

1. Navigate to the Recipe Validity Rules window.
2. Create or query a recipe. Enter as many parameters as needed to complete the search.
3. Select View Master Batch Record from Actions.
4. Print the Master Batch Record.

Viewing the Production Status

View current production status for multiple batches at a time. Navigate to the Production Status window from the MES for Process Manufacturing Administrator responsibility.

Viewing the Batch Progression

View detail batch progression details for a specific production batch.

To view the batch progression:

1. Navigate to the Production Status window.
2. Create or query batches. Enter as many parameters as needed to complete the search.

Select Organization, Batch, Product, Status, From Date, and To Date.

3. Click a batch number.

Viewing and Printing the Control Batch Record

View current production status for multiple batches at a time and also view detail batch progression details for a specific production batch.

Prerequisite

- ❑ In order to view the output of events modified by MES for Process Manufacturing in the Control Batch Record, verify that each Process Manufacturing transaction listed in the following table uses the style sheet listed in the same row as the transaction. For a list of modified events, see E-Record Events, page 9-5.

E-Record Transactions and Their Corresponding Style Sheets

Transaction Name	Transaction Type ID	E-Record Style Sheet
GME ERES Batch Hold Release	oracle.apps.gme.batch.holdrelease	GMEBHREL.rtf
GME ERES Batch Hold	oracle.apps.gme.batch.hold	GMEBHOLD.rtf
GME ERES Batch Fixed Process Loss Applied	oracle.apps.gme.batch.fixedprocessloss	GMEBEVRT.rtf
GME ERES Batch Step Activity Resource Process Parameters	oracle.apps.gme.batch.pparam	GMEBEVRT.rtf
GME ERES Batch Step Activity Resource Updated	oracle.apps.gme.resource.update	GMEBEVRT.rtf
GME ERES Batch Step Activity Updated	oracle.apps.gme.activity.updated	GMEBEVRT.rtf
GME ERES Batch Step Activity Removed	oracle.apps.gme.activity.removed	GMEBEVRT.rtf
GME ERES Batch Step Activity Added	oracle.apps.gme.activity.added	GMEBEVRT.rtf
GME ERES Batch Scaled	oracle.apps.gme.batch.scale	GMEBEVRT.rtf
GME ERES Batch Updated	oracle.apps.gme.batch.update	GMEBEVRT.rtf
GME ERES Batch Step Updated	oracle.apps.gme.batchstep.update	GMEBEVRT.rtf

Transaction Name	Transaction Type ID	E-Record Style Sheet
GME ERES Resource Transactions	oracle.apps.gme.batch.rsrc.trx	GMEBEVRT.rtf
GME ERES Batch Step Removed	oracle.apps.gme.batchstep.removed	GMEBEVRT.rtf
GME ERES Batch Step Added	oracle.apps.gme.batchstep.added	GMEBEVRT.rtf
GME ERES Batch Rerouted	oracle.apps.gme.batch.reroute	GMEBEVRT.rtf
GME ERES Batch Step Activity Resource Removed	oracle.apps.gme.resource.removed	GMEBEVRT.rtf
GME ERES Batch Step Activity Resource Added	oracle.apps.gme.resource.added	GMEBEVRT.rtf
GME ERES Batch Rescheduled	oracle.apps.gme.batch.rescheduled	GMEBEVRT.rtf
GME ERES Batch Material Removed	oracle.apps.gme.batchmtl.removed	GMEBEVRT.rtf
GME ERES Batch Material Updated	oracle.apps.gme.batchmtl.update	GMEBEVRT.rtf
GME ERES Batch Material Added	oracle.apps.gme.batchmtl.added	GMEBEVRT.rtf
GME ERES Batch E-Records	oracle.apps.gme.batch.erecords	GMEBEVRT.rtf
GME ERES Batch Canceled	oracle.apps.gme.batch.cancel	GMEBEVRT.rtf
GME ERES Batch Transact	oracle.apps.gme.batch.transact	GMEBEVRT.rtf
GME ERES Batch Step Complete	oracle.apps.gme.batchstep.complete	GMEBEVRT.rtf

Transaction Name	Transaction Type ID	E-Record Style Sheet
GME ERES Batch Complete	oracle.apps.gme.batch.complete	GMEBEVRT.rtf
GME ERES Batch Step Release	oracle.apps.gme.batchstep.release	GMEBEVRT.rtf
GME ERES Batch Step Close	oracle.apps.gme.batchstep.close	GMEBEVRT.rtf
GME ERES Batch Close	oracle.apps.gme.batch.close	GMEBEVRT.rtf
GME ERES Batch Release	oracle.apps.gme.batch.release	GMEBEVRT.rtf
GME ERES Batch Step Unrelease	oracle.apps.gme.batchstep.unrelease	GMEBEVRT.rtf
GME ERES Batch Step Revert to WIP	oracle.apps.gme.batchstep.uncertify	GMEBEVRT.rtf
GME ERES Batch Step reopen	oracle.apps.gme.batchstep.reopen	GMEBEVRT.rtf
GME ERES Batch Reopen	oracle.apps.gme.batch.reopen	GMEBEVRT.rtf
GME ERES Batch Unrelease	oracle.apps.gme.batch.unrelease	GMEBEVRT.rtf
GME ERES Batch Terminate	oracle.apps.gme.batch.terminate	GMEBEVRT.rtf
GME ERES Batch Revert to WIP	oracle.apps.gme.batch.uncertify	GMEBEVRT.rtf

If a transaction uses a different e-record style sheet than the style sheet listed in the preceding table, update the style sheet for that transaction.

To update the e-record style sheet, perform the following steps:

1. From the **Home** page, select the **ERES Administrator** responsibility, then **Administration Tasks**, and then **Setup**.

2. In the **Configuration Variables** page, search for and select a **Transaction Name**, then click **Go**.

The rows of associated transaction variables appear.

3. For the variable *E-record Style Sheet*, click **Update**.
4. In the **Update Transaction Variable** page, enter the E-record Style Sheet value.
5. Click **Apply**.

A Confirmation message appears.

To view and print the Control Batch Record:

1. Navigate to the Batch Progression window.
2. Click View Control Batch Record.

View or print the CBR from the Batch Details window using the Operator Workbench.

Control Batch Record (CBR) consists of:

- Material Details
 - Instructions
 - Samples
 - Nonconformance
 - Dispensing
 - Material Transactions
- Routing Steps with Overrides
 - Instructions
 - Nonconformance
 - Operations
 - Nonconformance
 - Instructions
 - Activities

- Nonconformance
- Instructions
- Resource Requirements
 - Instructions
 - Process Parameters
 - Resource Transactions
 - Nonconformance
- Heat Number
- Yield Details
 - Yield % for the batch. Yield % is displayed at batch complete and batch close events
 - Step transfer quantity for each step. This value is displayed at step complete, step close, batch complete, and batch close events
 - Yield % at step level. This value is displayed at step complete, step close, batch complete, and batch close events
- Attachments (Optional)

Batch Validation

This chapter covers the following topics:

- Requirement
- Understanding the Batch Validator
- Validating Batch Information

Requirement

Following is a requirement that customers have for validating the progress of their batches:

Can batch progress be validated?

Yes. Use the batch validator to verify the completeness of defined instructions and proper information entry.

Use the Batch Validator to Report Batch Progress

The batch validator verifies task completion and highlights exceptions or mismatched information. Use the batch validator to identify the completeness of information entry and proper authority to perform various tasks in the batch. Identify exceptions rapidly for appropriate corrective action or nonconformance reporting.

Understanding the Batch Validator

The batch validator highlights any exceptions, incomplete tasks, mismatch in data, and missing information. The key features of the batch validator include:

- Runtime Validation
- Reporting

Runtime Validation

Complete instructions for a batch after proper verification and identify any pending instructions at all levels of a batch. Verify and mark any pending or non-disposed nonconformance against a batch. The batch validator lets you make a secured disposition on the instructions. For example, only a supervisor can dispose instructions that are pending for a batch containing hazardous materials.

Reporting

The batch validator provides a paper-based reporting solution to help print the valuation output.

Validating Batch Information

The batch validator is triggered when you try to close a batch step or the batch itself from the Oracle Process Manufacturing Process Execution application. The batch validator ensures that all the tasks are performed as stipulated in the master batch record.

To validate batch automatically from batch step close or batch close:

1. Navigate to the Batch window.
2. Enter as many parameters as needed to complete the search. Select Recipe, Formula, Routing, or Product with appropriate Version for Recipe, Formula, or Routing. Select Organization, From Date, To Date, and Status.
3. Select a batch step or batch.
4. Click Close. The batch validator performs validations. A message displays if there are any exceptions that require corrective action.
5. Click Cancel to stop the closure of batch step or batch or click Skip to close the batch step or the batch.

Using the Touch Screen User Interface

Overview: Touch Screen User Interface

Oracle Process Manufacturing provides support for touch screen devices. A touch screen is an input device that enables you to touch the display screen. Shop floor personnel often need a user interface that requires minimal training and can be used in an environment where keyboards and pointing devices are not available.

You can use the touchscreen interface as a front end to the Operator Workbench to perform batch transactions, execute clearance and process instructions, or to perform dispensing and reverse dispensing operations. The touch screen application supports use of Dispense Areas and Dispensing Booths.

Related Topics

Creating Touch Screen Preferences, page 11-1

Creating Touch Screen Preferences

You create preferences from the Touchscreen Preferences page. This page lets you create, update, delete, or duplicate existing Touchscreen Preferences. You can define preferences at the following levels:

- User
- Responsibility
- Site

Preferences defined for a unique user and organization combination take precedence over preferences defined for a site or responsibility. So, if touchscreen preferences exist for a user and organization, then any preferences that exist for a responsibility or site and the same organization are ignored for that user. Although every touchscreen

preference must have an organization, only one preference at each unique level name and level value combination can have an assigned default organization.

To view and update touch screen preferences:

1. Navigate to the Touchscreen Preferences page.
2. Select Responsibility, Organization, User, or Site—and choose Go.

A summary of the preference records appear in a table per level, site, and organization.

Search Touch Screen Preferences Page

Level Name	Level Value	Organization	Default	Details	Duplicate	Update	Delete
Site	Yes	PR1	Yes				
User	PROCESS_OPS	PR1	No				
User	PROCESS_OPS	PR2	Yes				
Site	Yes	PR2	No				

The following information appears:

- Level Name. Responsibility, Site, or User.
 - Level Value. Displays Yes for Responsibility or Site. Displays the User Display Name when the User level is selected.
 - Organization. Required.
 - Default. Only one preference for each level name and level value combination can have a default organization selected.
 - Details icon
3. Select Details to view information created for a specific preference. The details appear in the following regions:
 - Preferences Level for Site, Responsibility, or User

- Operator Workbench
 - Dispensing
4. Select a record and choose Update to change or update values in a particular preference.

Touchscreen Preferences Page

Preferences Level			
	Level	User	Organization
	User	PROCESS_OPS	PR1
☒ Default Organization			
Operator Workbench			
Query Time Range	Week		Default Resource
Rolling	Yes		
Dispensing			
Dispense			
Dispense Area	PHAR-RM		Mode
Query Time Range	Week		Pallet Label
Rolling	Yes		Dispense Label
			Material Label
Scale			
Source	SCALE2		Full Mode Default Device
Target	SCALE2		Include Tare
Reverse Dispense			
Copy Dispense			
Dispense Area	PHAR-RM		Mode
Query Time Range	Week		Dispense Label
Rolling	Yes		Material Label
Scale			
Source	SCALE2		Full Mode Default Device
Target	SCALE2		Include Tare

To create touch screen preferences:

1. Navigate to the Touch Screen Preferences page and select Create.
2. In the Level field, select one of the following touch screen preference levels:
 - User
 - Responsibility
 - Site

3. Conditional. If you selected the Responsibility level, search for and select a responsibility. If you selected the User level, search for and select a user.
4. In the Organization field, search for and select an organization.
5. Optional. Select Default Organization. By selecting this option, you choose to use this organization as your default organization in the following pages:
 - Touchscreen Workbench
 - Touchscreen Dispensed Material Activity
 - Touchscreen Dispensable Material
 - Touchscreen Dispense Clearance
 - Touchscreen Print Labels

Additional Information: You can create many combinations of preferences. Individual user preferences override any applicable Responsibility preferences and Responsibility preferences override applicable Site preferences. Although many preferences at different levels may be valid for a user and organization combination, only one preference can be applied.

Only one preference at each unique level name and level value combination can have an assigned default organization.

6. In the Operator Workbench region, select the following time and resource values:
 - Query Time Range: Month, Number of Days, Week, or Today
 - Rolling: Yes or No
 - Default Resource
7. In the Dispensing region set preferences for Dispense and Reverse Dispense, as required. In the Dispense section, select the following values:
 - Dispense Area
 - Query Time Range: Month, Number of Days, Week, or Today
 - Rolling: Yes or No
 - Mode: Full Container, Source Container, or Target Container

- Pallet Label: Yes or No
 - Dispense Label: Yes or No
 - Material Label: Yes or No
8. In the Scale area, select the following:
- Source
 - Target
 - Full Mode Default Device
 - Include Tare: Yes, No, or null
 - Yes: Includes the tare weight in the quantity dispensed calculation (Quantity Dispensed = Quantity - Tare) by default. The Include Tare button is displayed in the Material Dispense: (item) page.
 - No: Excludes the tare weight by default in the quantity dispensed calculation (Quantity Dispensed = Quantity)
The Exclude Tare button is displayed in the Material Dispense: (item) page.
 - Null: Same as No.
- Use the Include/Exclude Tare buttons to switch between the two types of quantity dispensed calculations.
9. In the Reverse Dispense region, you can copy the data you entered in the Dispense region, or enter new values in the following fields:
- Dispense Area
 - Query Time Range: Month, Number of Days, Week, or Today
 - Rolling: Yes or No
 - Mode: Full Container, Source Container, or Target Container
 - Pallet Label: Yes or No
 - Dispense Label: Yes or No
 - Material Label: Yes or No
10. In the Scale area, select the following:

- Source
- Target
- Full Mode Default Device
- Include Tare: Yes, No, or null
 - Yes: Includes the tare weight in the quantity reverse dispensed calculation ($\text{Quantity Reverse Dispensed} = \text{Quantity} - \text{Tare}$) by default. The Include Tare button is displayed in the Reverse Dispense Material: (item) page.
 - No: Excludes the tare weight by default in the quantity reverse dispensed calculation ($\text{Quantity Reverse Dispensed} = \text{Quantity}$)
 The Exclude Tare button is displayed in the Reverse Dispense Material: (item) page.
 - Null: Same as No.

Use the Include/Exclude Tare buttons to switch between the two types of quantity reverse dispensed calculations.

11. Select Apply to save your work.

12. You also have the option to create a new record by selecting a record on the Search Touch Screen Preferences table and choosing Duplicate.

A copy of this record opens in the Touch Screen Preference Details page. You then change the Level Name, Level Value, or Organization values—and any other values before saving the record.

Setting Up and Using Dispense Area and Dispensing Booths

Setting up a Dispense Area and Dispensing Booths

Oracle MES for Process Manufacturing lets you re-create a true systematic representation of how dispensing happens on the shop floor. You can identify specific areas for inbound and outbound material storage. A Dispense Area represents the physical area in a warehouse where a dispensing activity occurs. The Dispensing Area can be divided into smaller rooms called Dispensing Booths which are the specific locations for dispensing activities. You can monitor environmental conditions at the Dispensing Booth level. A Dispense Area does not require a Dispensing Booth. You may create more than one Dispensing Booth for each Dispense Area.

Setting up a model for representing your actual dispensing operation allows rapid monitoring of inventory material movement in and out of the Dispensing Booth. You can identify which Operators access these areas, and monitor other specifications and environmental conditions. Using these tools you can meet regulatory requirements and assure better compliance.

To set up a Dispense Area, you first associate it with an organization, and then you define one or more Dispensing Booths, Storage Locations, and Area Human Resources.

To set up a Dispense Area:

1. Navigate to the Dispense Area window.
2. Click Create to create a new Dispense Area or use the Search region to select an existing Dispense Area.
3. Enter an Organization. Required.
4. Enter a name for the Dispense Area. Required.

5. Enter a Description for the Dispense Area. Required.
6. Enter a Subinventory for the Dispense Area. Required.
7. Enter a Max Number of Tasks per Day for the Dispense Area. Required.
8. In the Instruction Set region, optionally create a dispense clearance instruction set. See: Clearing Dispense Areas, page 5-6.
9. In the Dispensing Booths region, create one or more Dispensing Booths. Click Add Another Row to create more than one Dispensing Booths.
10. In the Storage Locations region, create one or more Storage Locations. Click Add Another Row to create more than one Storage Locations.
11. In the Area Human Resources region, create one or more Area Human Resources. Click Add Another Row to create more than one Area Human Resource.
12. Click Apply.

Clearing a Dispense Area Using the Touch Screen Interface

Your production process may require that you clear the dispense area before starting the dispense process for a batch. Examples of tasks associated with clearing a dispense area include:

- Cleaning
- Calibrating equipment
- Recording dispense booth temperature and humidity conditions

Acknowledge and record dispense area clearance tasks using the touch screen interface as described below. Alternatively, execute dispense clearance instructions as described in Clearing Dispense Areas, page 5-6.

To clear a dispense area using the touch screen interface:

1. Navigate to the **Touchscreen Dispense Clearance** page.
2. Select an **Organization** from the list. Required.
3. Select a **Dispense Area** from the list.
4. Touch, tap, or press the **Accept** check mark to continue.
The list of batches with pending dispensable materials appears.

5. Touch, tap, or press **Dispense Clearance** for a pending dispensable material to continue.

The **Dispense Material Instructions** page for the material opens, listing all clearance instructions for the dispense area.

6. Touch, tap, or press:
 - **Comments** to enter any additional information
 - **Attributes**, if enabled, to enter any required or optional instruction attributes.
 - **Attachments** to optionally attach a file, URL, long text, or short text.
7. Acknowledge each instruction. Touch, tap, or press **Acknowledge** to continue.
8. After acknowledging all instructions, touch, tap, or press the **Accept** check mark to continue.

The system returns you to the **Dispense Material Instructions** page where a message appears, confirming that you cleared the dispense area.

Related Topics

Clearing Dispense Areas, page 5-6

Dispensing Using the Touch Screen Interface

You can dispense materials using the Touchscreen Dispensable Material window. This functionality offers the same options for dispensing as the Dispensable Material window.

To dispense using the touch screen user interface:

1. Navigate to the Touchscreen Dispensable Material window.
2. Select an Organization from the list. Required.
3. Select a Dispense Area from the list.
4. Touch, tap, or press the Accept check mark to continue.
5. In the Work List window, touch, tap, or press Reset or Change to alter the Search Criteria.

Additional Information: You can search for material to dispense by

selecting a Dispense Area, Date Range, Batch, Item, or Lot. You can also touch, tap, or press the Organization Name to select a new organization.

6. Locate the material to dispense and touch, tap, or press the corresponding Navigate to Task button in the Dispense column.

Tip: Use the First, Previous, Next, and Last arrows to page through the Work List.

7. Review the information in the Dispense Material Instructions window. Select Show Details to see all available information.
8. In the Instruction Set region, optionally create a dispense clearance instruction set. See: Clearing Dispense Areas, page 5-6.
9. In both the Clearance and Dispense tabs, touch, tap, or press:
 - Comments to enter any additional information
 - Attributes (in the Dispense tab only), if enabled, to enter any required or optional instruction attributes.
 - Attachments to optionally attach a file, URL, long text, or short text.
10. Touch, tap, or press Acknowledge to acknowledge that you have performed mandatory instructions or have reviewed, at a minimum, optional instructions.
11. Touch, tap or press Accept to continue or Cancel to return to the Work List without dispensing the material.
12. In the Material Dispense window enter a Quantity by selecting the Number Pad.
13. Change the Mode, Selection, or Security as appropriate.
14. Select a Booth. Required.
15. Touch, tap, or press Continue.
16. In the Review window, review the information and then Touch, tap, or press Save to dispense the material.

Reverse Dispensing Using the Touch Screen Interface

You can reverse dispense materials using the Touchscreen Dispensed Material Activity window. This functionality offers the same options for dispensing as the Dispensed Material Activity window.

To reverse dispense using the touch screen user interface:

1. Navigate to the Touchscreen Dispensed Material Activity window.
2. Select an Organization from the list. Required.
3. Select a Dispense Area from the list.
4. Touch, tap, or press the Accept check mark to continue.
5. In the Work List window, touch, tap, or press Reset or Change to alter the Search Criteria.

Additional Information: You can search for material to dispense by selecting a Dispense Area, Date Range, Batch, Item, or Lot. You can also touch, tap, or press the Organization Name to select a new organization.

6. Locate the material you want to reverse dispense and touch, tap, or press the corresponding Navigate to Task button in the Reverse Dispense column.

Tip: Use the First, Previous, Next, and Last arrows to page through the WorkList.

7. In the Reverse Dispense Material window enter the Quantity of Material Loss and the Security Tag. Required.
8. Review the information in this window, make any necessary changes, and touch, tap, or press Continue.
9. In the Review window, review the information and then Touch, tap, or press Save to reverse dispense the material.

Setting Up and Using Touch Screen Operator Workbench

Overview: Touch Screen Operator Workbench

You use the Operator Workbench's touch screen interface to perform batch transactions and execute clearance and process instructions. The Worklist window lists all of the operations of the Batch for the Organization that you can access. You can see the details of each operation in the Worklist window. You can also change the contents of the Worklist window by specifying new search criteria. The Worklist window can take you to the instructions where you can see the SOP's, and respond to the individual tasks. Operators must complete mandatory tasks in the proper order.

The key functions supported by the Touchscreen Operator Workbench are:

- Identify the tasks to complete.
- View pending instructions.
- Execute process instructions.
- Release and complete batch step.

Note: Before performing various batch transactions and executing any process instructions, you must select an appropriate organization from the list on the Choose Organization page.

Using the Touch Screen Operator Workbench

The Operator Workbench: Worklist page displays a list of batches that are in various statuses such as WIP, Pending, Completed, Closed, and Pending for the selected organization. The worklist page enables you to execute process instructions for the

batches and batch steps that are in WIP and Pending status. You can use the Operator Workbench: Worklist page to view details and SOPs associated with the process instructions that are defined in the Recipe window (Oracle Process Manufacturing Product Development) and must be followed by the production operators at the time of executing the batch. You can also view the pending instructions and execute the instructions to complete a batch step.

The worklist page displays the following details:

Column Name	Description
Batch	Displays the batch number.
Step	Displays the batch step number.
Operation	Displays the name of the operation.
Product	Displays the name of the product yielded by the operation.
Material Availability	Displays availability of the required materials.
Planned Start Date	Displays planned start date for the batch.
Planned Completion Date	Displays planned completion date for the batch.
Lock Owner	Displays the name of the user if the batch is locked.
Details	Use this icon to view and execute process instructions and also to view associated SOPs.
Execute	Displays options for batch progression. The available options are: • Release: Used for releasing batch step and executing process instructions. This button is used for changing the status of a batch from Pending or WIP to Release. • Complete: Used for completing the batch step after executing the process instructions and changing status from Release to Complete.

Searching for Batches

You can use the arrow buttons to navigate to next pages of the batch list or you can use the Search functionality to find and view a specific batch.

To search for a batch:

1. Navigate to the Touchscreen Workbench.
2. Select the appropriate organization from the Organization list.
3. Touch, tap or press Accept to open the Operator Workbench: Worklist page.
4. In the Search Criteria region, touch, tap or press the Change button.
5. In the Select Search Criteria page, use any of the following criterion to refine your search:
 - Resource
 - Date Range
 - Batch
 - Product
 - Operation
 - Batch Step Status
6. Touch, tap or press Accept to view the search results.

Viewing Pending Instructions

The Process Instructions page enables you to view the instructions for the selected batch step. The process instructions must be executed in order to complete the operation and progress the batch.

Use the touch screen device to view the following the process instructions:

- All: Displays all pending and mandatory instructions.
- Pending Instructions: Displays mandatory and optional pending instructions.
- Mandatory Pending Instructions: Displays all pending instructions that must be executed to complete the operation.

To view the Pending Instructions using the touch screen:

1. Navigate to the Operator Workbench: Worklist page.

2. Search for a batch in the WIP or Pending status.
3. Touch, tap or press the Details icon to view process instructions for a batch.
4. Touch, tap or press Execute Instructions to open the Process Instructions window.
5. In the Process Instructions page, select All, Pending, or Mandatory Pending instructions from the Show list.

After you select an option from the Show list, the screen refreshes to only show the selected type of process instructions.

Executing Process Instructions Using the Touch Screen Operator Workbench

Process Instructions explain the sequenced set of activities and directions for production operators to follow when progressing the batch step. The process instructions must be executed to complete the operation and progress the batch.

You can execute the following process instructions using the Operator Workbench: Worklist page:

- Material Transactions
- Resource Transactions
- Process Parameters
- Quality Samples
- Quality Results

Prerequisites:

1. Create Instructions
2. Change batch status to WIP or Pending

To execute Process Instructions using the touch screen:

1. Navigate to the Operator Workbench: Worklist page.
2. Search for and select a batch in the WIP or Pending status, and then click Release.
3. Touch, tap or press the Details icon to view process instructions for the selected batch.
4. For Process Parameters and Resource Transactions only, touch, tap or press an SOP under the SOP column to view SOP details.
5. Touch, tap or press Execute Instructions to open the Process Instructions window.

6. Optionally, touch, tap or press:
 - Comments to enter any additional information
 - Attributes, if enabled, to enter any required or optional instruction attributes.
 - Attachments to optionally attach a file, URL, long text, or short text.

Required Action: Touch, tap or press Acknowledge in the Response column for all mandatory pending instructions. You can not respond to other instructions until after you complete the mandatory instructions. Use the Show list to view pending, mandatory pending, or all instructions.

Conditional Action: If you choose not to complete every pending optional instruction, you have the option to click Mark Instruction Set as Complete. This marks the instruction set with a status of Complete, but gives the incomplete optional instructions a status of Excluded. Once optional instructions are excluded, you no longer have the option to execute them. Alternatively, you can click Mark All Instructions as Complete. This marks each optional instruction as complete as well as the instruction set as complete. The Mark All Instructions as Complete option is not available if one or more of the pending optional instructions requires an e-record and e-signature approval.

7. In the Process Instructions window, touch, tap, or press one of the following buttons in the Response column to open the similarly named window for the instruction to process:
 - Material Transaction
 - Resource Transaction
 - Process Parameter
 - Quality Touch Screen Sample
 - Quality Touch Screen Result
8. Next, follow the steps listed below for executing process instructions for different types of transactions.

Creating Material Transactions Using the Touch Screen Operator Workbench

When executing process instructions for a batch operation, you must approve material transactions for each material used in the operation. Begin this process from the Material Transactions window.

To create material transactions using the touch screen operator workbench:

1. In the Material Transactions window, touch, tap or press Select to include the materials to process according to the instruction.
2. Touch, tap or press Accept to include the selected materials.

Conditional Action: If the materials are lot, location, and/or serial-controlled, perform these steps to select materials:

1. Touch, tap, or press Add Another Row.
 2. In the Subinventory field, select the subinventory from which to transact the material quantity.
 3. If the item is locator-controlled, in the Locator field, enter or select the item's locator.
 4. If the item is lot-controlled, in the Lot field, enter the item lot or select a lot from the list of available lots.
 5. In the Quantity field, enter the quantity of the item to transact.
 6. If the item is serial-controlled, touch, tap, or press Serials.
 7. In the Serials page, touch, tap, or press Add Rows.
 8. In the From Serial Number field, search for and select a starting serial number for the quantity to transact.
 9. In the To Serial Number field, search for and select an ending serial number.

Touch, tap, or press Add Rows to enter more From/To Serial Number ranges, if needed, for the quantity to transact.
 10. Touch, tap, or press Accept to accept the serial numbers.
3. If you are using Oracle E-Signatures, in the E-Signatures window, then enter your User ID and Password and touch, tap, or press Approve to approve the materials. Press Accept again to complete the operation.

4. In the Operation Workbench: Worklist page; touch, tap or press Complete to change the status of batch from Release to Complete.

Creating Resource Transactions Using the Touch Screen Operator Workbench

When executing process instructions for a batch operation, you must perform resource transactions for each resource used in the operation. You begin this process from the Resources window.

To create resource transactions using the touch screen operator workbench:

1. In the Resources window, touch, tap, or press Perform to open the Resource Transactions window for the resource you want to use.
2. Review and update the Start Date, End Date, Usage, and Reason for the resource transaction, as required.
3. Update the Flex Field details for each resource transaction, if required.
4. Touch, tap or press Accept to include the selected resource transactions in the batch operation.
5. If you are using Oracle E-Signatures, in the E-Signatures window, enter your User ID and Password. Touch, tap, or press Approve to approve the resource transactions. Press Accept again to continue.
6. Touch, tap, or press Execute Instructions in the Batch Operation window to complete the operation.
7. In the Operation Workbench: Worklist page; touch, tap or press Complete to change the status of batch from Release to Complete.

Setting the Process Parameters for Process Instructions Using Touch Screen Operator Workbench

When executing process instructions for a batch operation, you must enter process parameters for each device in the operation. You need to do this for each processing instruction to execute.

To set process parameters for instructions:

1. In the Process Parameters page, touch, tap or press Select to include the parameters to apply to the instruction.
2. To modify the individual parameters, touch, tap or press the Keypad to change the Actual Value of the parameter.

- You can select **Read Device** to update the information for each parameter.
 - If multiple readings are enabled for the process parameter, touch, tap or press Add Another Row. Enter one reading date and reading value in each row. Add as many rows as needed. After all readings are entered, touch, tap or press Calculate Aggregates to view the aggregated values of all readings entered.
3. Update the Flex Field details for each parameter, if required.
 4. Touch, tap, or press Accept.
 5. If you are using Oracle E-Signatures, in the E-Signatures window, enter your User ID and Password. Touch, tap, or press Accept to approve the changes. Press Accept again to continue.
 6. In the Operation Workbench: Worklist page; touch, tap or press Complete to change the status of batch from Release to Complete.

Creating Quality Samples Using Touch Screen Operator Workbench

The Sample Creation window enables you to create quality samples for a particular batch step.

To create a quality sample using the touch screen:

1. In the Sample Creation window, search for and select the following sample information:
 - Spec Name: Optional.
 - Sample: Enter the sample number if document numbering is manual. If document numbering is automatic, then this field is disabled.
 - Parent Lot: Mandatory, but only if delay lot entry is not allowed.
 - Lot: Mandatory, but only if delay lot entry is not allowed.
 - LPN: Optional
 - Quality Laboratory: Mandatory. Select a laboratory type organization from the list of values.

Optionally, enter a description of the sample, then select the following dates using the Pick-a-Date page for each of the following dates:

- Drawn Date: Defaults to the system date.
- Received Date
- Required Date

Optionally, select Label Print to print a label for the sample.

Touch, tap or press Accept to create the sample.

2. Touch, tap or press Accept to save your quality samples until you return to the Operator Workbench: Worklist page.

A message indicating either successful creation of the sample or error appears. If a sample is created, it will have a sample disposition of Pending.

Entering Quality Results Using the Touch Screen Operator Workbench

The Samples Collected window opens if more than one sample is collected for the batch step. If only one sample is collected for the step, then the Result Entry window opens for the sample.

Prerequisites for Entering Quality Results with Operator Certification Checks

You can associate defined competencies to persons designated as testers who are qualified to operate a set of test equipment and perform the tests. If a test requires a certain competency and proficiency level, perform the following prerequisites before using the Touch Screen Operator Workbench to enter quality results.

1. Perform the prerequisite steps listed in Entering Results with Operator Certification Checks, *Oracle Process Manufacturing Quality Management User's Guide*.
2. Add a process instruction to the recipe operation step from where you take a product quality sample.
 1. In the Recipe Details window, select the recipe operation step from where you take a product quality sample.
 2. Click Step/Material Association.
 3. In the Step/Material Association window, associate the operation step to the product.
Close the Step/Material Association window.
 4. In the Recipe Details window, select the Actions menu, then Process Instructions.
 5. Create a process instruction set if one does not already exist.

6. In the new or existing process instruction set, create an instruction as described in Creating Instructions, page 3-7, but select the following field values:
 - Acknowledgement: Mandatory
 - Task: Quality Touch Screen Result
 - Task Attribute: the product quality sample item
3. Create a WIP specification for the product and batch as described in Entering an Item Specification Validity Rule, *Oracle Process Manufacturing Quality Management User's Guide*.

To enter a quality result using the touch screen:

1. In the Samples Collected window, select the sample number for which to enter results.
Touch, tap or press Results.
2. To add a test performed on the sample, select one of the following options from the Actions list, then tap Go.
 - Add Test: Specify the class, test, test quantity, and test quantity unit of measure. Tap Accept to add the test.
 - Add Test Groups: Enables you to select multiple tests within a test class at one time. Select a test class, then select which tests within the test class to add. Enter the test quantity and test quantity unit of measure for each test. Tap Accept to add the tests.
3. For each test performed, the following field values are defaulted based on the sample's specification. Change to actual values, as needed.
 - Laboratory
 - Device: The device used to perform the test.
 - Result: The test's target result.
 - Test Unit: The test's unit of measure.
 - Evaluation: The value defaults based on the Result value.
 - Min: The test's acceptable minimum result, if applicable.
 - Max: The test's acceptable maximum result, if applicable.

4. Select the appropriate action, if necessary, from the Action list for each test.
5. If a test is performed by a device, select the test.

Tip: If the same device performs multiple tests, select all tests performed by the device.

6. Touch, tap, or press Read Device to retrieve the test's actual result from the device.
7. Touch, tap, or press Accept to save your quality results until you return to the Operator Workbench: Worklist page.

Conditional Action: If the test is associated to a test method that requires an operator certification above your proficiency level to enter results and the test method permits overrides, then a qualified person must give approval for you to enter results. A Login page appears so the qualified person can grant permission for results entry.

If the test method does not permit overrides, then an error message appears to inform you that you are not qualified to perform this transaction.

For more information, refer to Entering Results with Operator Certification Checks, *Oracle Process Manufacturing Quality Management User's Guide*.

A message indicating either successful entry of the result or error appears.

Using the Dispense Supervisor Workbench

Viewing and Updating Dispense Task Assignments

The Dispense Supervisor Workbench helps you plan and manage dispensing tasks. It provides a Dispense Supervisor with the capability to manipulate dispensing tasks prior to processing by the Dispense Operator. The Dispense Supervisor uses this workbench to sort and refine the list of pending dispense tasks available to the Dispense Operator.

You can use the Dispense Supervisor Workbench to group dispense tasks together to create a master dispense plan. You can change the dispense required date to improve the facilitation of batch processing, and assign a priority to each dispense task. Change or assign Lot Numbers and Dispense Areas, and make use of the Mass Update feature to change several dispensing tasks simultaneously. Assign Dispense Operators to the specific tasks for which they are best suited. By being able to control what you are dispensing, you avoid dispense expiration - the maximum acceptable duration between dispensing and consumption of the dispensed batch material. The Dispense Supervisor Workbench displays the Required Date and the Dispense Expiration date for each dispense task. If a Dispense Expiration date has elapsed, then the Dispense Supervisor Workbench highlights the task in red, and warns you with a message prior to material consumption.

Use the Dispense Supervisor Workbench to view current dispense task assignments for your organization. When looking at individual dispense task assignments, you can view them in either text or graphical mode by clicking either Toggle to Text Mode or Toggle to Graphic Mode.

To view and update dispense task assignments:

1. Navigate to the Dispense Supervisor Workbench window.
2. Select an Organization from the list. Required.
3. Click **View Current Assignments** to display all the assigned dispense tasks for the

organization.

4. Enter the criteria to query in the Search region.
5. Click Go to narrow the list of current assignments.
6. Click Select to mark individual assignments for Mass Update or Reverse Dispense.
7. Click Update to open the assignments to view or edit.
8. Update the appropriate information in various fields such as Dispense Area, Sub-inventory, Quantity, Planned Dispense Date, and Operator etc., and click Apply. For example, to change the operator, enter the name of the person in the Operator field and click Apply.

To find unplanned dispense tasks:

1. Navigate to the Dispense Supervisor Workbench window. Select an Organization from the list. Required.
2. In the Search region, select Unplanned from the Status list. Click Go.
3. Use the Update region to edit the unplanned dispense tasks for the assignment.

Note: Use the Dispense Supervisor Workbench Current Dispensing Task Load window to perform dispense planning for these unplanned dispense tasks.

4. Click Apply.

Setting Up and Using Operator Certification

Understanding Operator Certification

You can use Operator Certification to ensure that only certified operators perform tasks in your implementation of MES for Process Manufacturing. By defining and enforcing certifications, proficiencies, and competencies, you can help meet the regulatory requirements of the FDA 21 CFR Part 211, and comply with other regulatory requirements such as OSHA. Use Operator Certification to define which certification, competency, and proficiency each task in MES for Process Manufacturing requires. Operator Certification validates each task during execution, and allows you to override the requirements.

Operator Certification helps assure that only certified operators perform critical tasks, and eliminates many errors that result from insufficient training or competency. Each time an operator performs a transaction, Operator Certification assures that the operator is certified to do so.

Operator Certification System Dependencies

Operator Certification in MES for Process Manufacturing requires that you implement portions of Oracle HRMS. Oracle HRMS stores information pertaining to each employee who uses MES for Process Manufacturing. You define each certification, competency, and proficiency in Oracle HRMS, and track these three elements for each employee in the HRMS application. Using Oracle Learning Management, a Learning Administrator can reference each certification and competency within a training course, and then each Employee can complete the required certification using Oracle Self-Service Learning Management.

To use Oracle HRMS to enter employee data, refer to the *Oracle HRMS Enterprise and Workforce Management Guide*.

To use Oracle HRMS to define each certification, competency, and proficiency, refer to *Defining Competencies, Oracle HRMS Workforce Sourcing, Deployment, and Talent Management Guide* and *Creating a Competency Profile, Oracle HRMS Workforce Sourcing,*

Deployment, and Talent Management Guide.

To use Oracle Learning Management to reference each certification and competency within a training course, refer to the *Oracle Learning Management User Guide*.

Associating a Certificate or Competency with an Object Type

To set up Operator Certification in Oracle MES for Process Manufacturing, you use the Operator Certification - Create Object window to associate Items, Resources, Test Methods, and Transactions with Certificates and Competencies. If you set up a certification requirement for an Item or Resource, and for a Transaction using that Item or Resource, then before the operator performs the action, be aware that the certification will have to be verified twice.

Additional Information: In addition to the steps below, you must also enable the profile option "GMO:Operator Certificate for Process Manufacturing is enable" in order to require operator certification. See: Setting Up Profile Options, page 2-1.

To associate a certificate or competency with an Object Type:

1. Navigate to the Operator Certification-Create Object window.
2. Enter or search for and select an Organization. Required.

Important: When associating a certificate or competency with a test method, enter a laboratory organization for the test method.
3. Select either Item, Resource, Test Method or Transaction from the Object Type list. Required.
4. Click Override to override the certification or competence.
5. Select an Object Type, then select a specific object (an Item, Resource, Test Method or Transaction) from the list.
6. Click Go to open up a list of existing objects, or click Create to create a new one.
7. In the Competence Criteria region select either Certification or Competency from the Type list.
8. Select the appropriate Certificate or Competence for the Item, Resource, Test Method or Transaction. Required

Note: Certificates and Competencies must exist in Oracle HRMS in order to be selectable here.

9. Enter or search for a Proficiency rating.
10. Enter a Start Date for the certification or competency. Required.
You can also enter an End Date for the competency.
11. Click Apply.

Overriding a Certificate or Competency Requirement

During dispensing, if an operator does not have the required certificate or competency to manage the item, then a qualified person must give approval for the operation to continue. Only certain requirements are eligible for override. If a certificate cannot be overridden, then an error message explains that the competency or certificate is mandatory.

To override a certificate or competency requirement

1. Enter your Username. Required.
2. Enter your Password. Required.
3. Enter any necessary Comments.
4. Click Override.

MES for Process Manufacturing Navigation Paths

Oracle MES for Process Manufacturing Navigation Paths

Although your System Administrator may have customized your navigator, typical navigation paths are described in the following table. In some cases, there is more than one way to navigate to a window, page, or Touchscreen interface. This table provides the most typical default path.

(T) indicates a tab, (M) indicates a menu, (L) indicates a link, and (B) indicates a button.

Window	Path
Add Instructions	Instructions > Add Existing
Batch Progression	Production Status > Search > Batch Details
Control Batch Record	Batch Progression > View Control Batch Record
Create Dispensing Setup	Dispensing Setup > Create
Create Document Number	Document Numbering > Create
Create Instruction	Follow the navigation path for Instruction Set. Click the instruction set name. Click Create.
Create Instruction Set	Follow the navigation path for Instruction Set. Click Create.

Window	Path
Dispense Clearance	Administrator > Dispensing > Dispense Clearance
Dispensable Material	Administrator > Dispensing > Dispensable Material
Dispensable Material: Worklist	Administrator > Dispensing > Touchscreen Dispensable Material
Dispense Area	Administrator > Setup > Dispense Area
Dispense Material	Dispense Material Instructions > Apply
Dispense Material Instructions	Administrator > Dispensing > Touchscreen Dispensable Material (B) Dispense
Dispensed Material Activity Details	Dispensed Material Activity > Search > Details
Dispense Material Instructions	Dispensable Material > Search > Dispense
Dispensed Material Activity	Administrator > Dispensing > Dispensed Activity
Dispensing Setup	Administrator > Setup > Dispensing Setup
Document Numbering	Administrator > Setup > Document Numbering

Window	Path
Instruction Set	Create Dispensing Setup > Create Instructions Process Engineer > Recipes > Recipe Details > Actions > Process Instructions Process Engineer > Process Routings > Process Routing Details > Actions > Process Instructions Process Engineer > MES Operations > Process Operation Details > Actions > Process Instructions Process Engineer > MES Operations > Process Operation Details > Activities > Actions > Process Instructions Process Engineer > MES Operations > Process Operation Details > Resources > Actions > Process Instructions
Instructions	Create Instruction Set > Apply > Instruction Set Name
Master Batch Record	Process Engineer > Engineering Workbench > Recipe Details
Operator Workbench	Administrator > Workbench
Operator Certification	Setup > Operator Certification
Process Instruction Attributes	MES for Process Manufacturing Administrator > Administrator: Workbench > Process Instruction Attributes
Production Status	MES for Process Manufacturing Administrator > Inquiry
Print Label	Administrator > Dispensing > Dispensable Material > Dispense > Search > Dispense > Apply > Dispense Material > Print Label Administrator > Workbench > Touchscreen Print Labels

Window	Path
Reverse Dispense Material	Reverse Dispense Material Instructions > Apply
Reverse Dispense Material Instructions	Dispensed Material Activity > Search > Reverse Dispense
Samples Collected	Administrator: Workbench > Touchscreen Workbench > Operator Workbench: Worklist (B) Details (for a batch or batch step) (T) Process Instructions (B) Execute Instructions (B) Quality Touch Screen Result
Update Dispensing Setup	Dispensing Setup > Search > Update
Update Document Number	Document Numbering > Search > Update
Update Instruction Set	Instruction Set > Update
Update Instructions	Instructions > Update

Oracle E-Record Events in Oracle MES for Process Manufacturing

This appendix covers the following topics:

- Oracle E-Record Events in MES for Process Manufacturing
- Events
- Subscriptions

Oracle E-Record Events in MES for Process Manufacturing

Oracle Manufacturing Execution System (MES) for Process Manufacturing supports the acquisition of electronic signatures on electronic records. Electronic documentation replaces paper with a seamless interface of business activities that improves manufacturing throughput and enhances the overall quality of operations in the supply chain. The following provides information for events that are e-record and e-signature enabled in the Oracle Manufacturing Execution System (MES) for Process Manufacturing application.

Events

The following tables list the events that are e-record and e-signature enabled and used by Oracle Manufacturing Execution System (MES) for Process Manufacturing. Refer to documentation of specific event details in other applications where indicated.

Event	Online or Deferred
GMD ERES Sample Creation Event	Refer to OPM Quality Management
GMD ERES Results Entry	Refer to OPM Quality Management

Event	Online or Deferred
Batch Close	Refer to OPM Process Execution
Batch Complete	Refer to OPM Process Execution
Batch Release	Refer to OPM Process Execution
Batch Reopen	Refer to OPM Process Execution
Batch Scale	Refer to OPM Process Execution
Batch Termination	Refer to OPM Process Execution
Batch Revert to WIP	Refer to OPM Process Execution
Batch Unrelease	Refer to OPM Process Execution
Batch Step Addition	Refer to OPM Process Execution
Batch Step Close	Refer to OPM Process Execution
Batch Step Complete	Refer to OPM Process Execution
Batch Step Release	Refer to OPM Process Execution
Batch Step Removal	Refer to OPM Process Execution
Batch Step Reopen	Refer to OPM Process Execution
Batch Step Revert to WIP	Refer to OPM Process Execution
Batch Step Unrelease	Refer to OPM Process Execution
Batch Step Update	Refer to OPM Process Execution
Resource Transaction	Refer to OPM Process Execution
Batch Update	Refer to OPM Process Execution
Batch Material Addition	Refer to OPM Process Execution

Event	Online or Deferred
Batch Material Removal	Refer to OPM Process Execution
Batch Material Update	Refer to OPM Process Execution
Batch Step Activity Resource Addition	Refer to OPM Process Execution
Batch Step Activity Resource Removal	Refer to OPM Process Execution
Batch Step Activity Resource Update	Refer to OPM Process Execution
Batch Transaction	Refer to OPM Process Execution
Batch Cancellation	Refer to OPM Process Execution
Batch Reschedule	Refer to OPM Process Execution
Batch Reroute	Refer to OPM Process Execution
Batch Creation	Refer to OPM Process Execution
Batch Step Activity Resource Process Parameters	Refer to OPM Process Execution
Batch Step Activity Update	Refer to OPM Process Execution
Batch Step Activity Removal	Refer to OPM Process Execution
Batch Step Activity Addition	Refer to OPM Process Execution
GMO ERES Material Reverse Dispense Event	Online
GMO Instruction Set Update	Online
GMO ERES Material Dispense Event	Online
GMO ERES Label Printing	Online
QA ERES Nonconformance Detail Approval	Refer to Oracle Quality
QA ERES Nonconformance Update	Refer to Oracle Quality

Event	Online or Deferred
QA ERES Nonconformance Creation	Refer to Oracle Quality
QA ERES Nonconformance Master Approval	Refer to Oracle Quality
QA ERES Disposition Creation	Refer to Oracle Quality
QA ERES Disposition Update	Refer to Oracle Quality
QA ERES Disposition Header Approval	Refer to Oracle Quality
QA ERES Disposition Detail Approval	Refer to Oracle Quality

Note: The GMO Instruction Set Update event (oracle.apps.gmo.instrset.update) is not an e-record or e-signature event. The event is raised at Instruction Set Level to include the instructions in Control Batch Record.

Oracle MES for Process Manufacturing Event Information

Name	Display Name	Description	Status	Owner Name	Owner Tag
oracle.apps.gmo.instruction.complete	GMO ERES Instruction Complete	GMO ERES Instruction Complete	ENABLED	MES for Process Manufacturing	GMO
oracle.apps.gmo.mtl.revdisp	GMO ERES Material Reverse Dispense Event	GMO ERES Material Reverse Dispensing Business Event	ENABLED	MES for Process Manufacturing	GMO
oracle.apps.gmo.labelprint	GMO ERES Label Printing	GMO ERES Label Printing Event	ENABLED	MES for Process Manufacturing	GMO

Name	Display Name	Description	Status	Owner Name	Owner Tag
oracle.apps.gmo.dispense.setup.create	GMO ERES Dispense Setup Creation Event	This event is raised for each dispense setup that is created.	ENABLED	MES for Process Manufacturing	GMO
oracle.apps.gmo.dispense.setup.update	GMO ERES Dispense Setup Update Event	This event is raised each time a dispense setup is updated.	ENABLED	MES for Process Manufacturing	GMO
oracle.apps.gmo.mtl.dispense	GMO ERES Material Dispense Event	MES for Process Manufacturing - ERES Material Dispense	ENABLED	MES for Process Manufacturing	GMO
oracle.apps.gmo.mbr.approve	GMO ERES Master Batch Record Approval	Master Batch Record Approval	ENABLED	MES for Process Manufacturing	GMO
oracle.apps.gmo.cbr.approve	GMO ERES Control Batch Record Approval	Control Batch Record Approval	ENABLED	MES for Process Manufacturing	GMO

Event Keys

Name	Event Key	User Event Key
GMO ERES Instruction Complete	INSTRUCTION_ID	INSTR_NUMBER
GMO ERES Material Reverse Dispense Event	UNDISPENSE_ID	UNDISPENSE_NUMBER

Name	Event Key	User Event Key
GMO ERES Label Printing	LABEL_ID	Business Flow Name Label
GMO ERES Dispense Setup Creation Event	CONFIG_ID	ITEM
GMO ERES Dispense Setup Update Event	CONFIG_ID	ITEM
GMO ERES Material Dispense Event	DISPENSE_ID	DISPENSE_NUMBER
GMO ERES Master Batch Record Approval	RECIPE_ID-ORG_ID- RECIPE_VALIDITY_RULE_I D	RECIPE_NO-VERSION (Org_Code)
GMO ERES Control Batch Record Approval	Batch_ID	BATCH_NO(Org_Code)

Subscriptions

Source Type	Event Filter	Phase	Rule Data	Rule Function	Parameters
Local	oracle.apps. gmo.cbr. approve	0	KEY	EDR_PSIG_R ULE. PSIG_RULE	EDR_AME_T RANSACTIO N_TYPE=orac le.apps.gmo. cbr.approve
Local	oracle.apps. gmo.mbr. approve	0	KEY	EDR_PSIG_R ULE. PSIG_RULE	EDR_AME_T RANSACTIO N_TYPE=orac le.apps.gmo. mbr.approve

Source Type	Event Filter	Phase	Rule Data	Rule Function	Parameters
Local	oracle.apps.gmo.instruction.complete	0	MESSAGE	EDR_PSIG_RULE. PSIG_RULE	EDR_COPY_ATTACHMENTS= Y or N EDR_AMENDMENT_TRANSACTION_TYPE=oracle.apps.gmo.instruction.complete EDR_XML_METADATA_CODE=gmoinsmp
Local	oracle.apps.gmo.labelprint	0	KEY	EDR_PSIG_RULE. PSIG_RULE	EDR_AMENDMENT_TRANSACTION_TYPE=oracle.apps.gmo.labelprint EDR_XML_METADATA_CODE=oracle.apps.gmo.labelprint FORCE_RECORD=Y
Local	oracle.apps.gmo.dispense.setup.create	0	KEY	EDR_PSIG_RULE. PSIG_RULE	EDR_AMENDMENT_TRANSACTION_TYPE=oracle.apps.gmo.dispense.setup.create
Local	oracle.apps.gmo.dispense.setup.update	0	KEY	EDR_PSIG_RULE. PSIG_RULE	EDR_AMENDMENT_TRANSACTION_TYPE=oracle.apps.gmo.dispense.setup.update

Source Type	Event Filter	Phase	Rule Data	Rule Function	Parameters
Local	oracle.apps. gmo.mtl. dispense	0	KEY	EDR_PSIG_RULE. PSIG_RULE	EDR_XML_MAP_CODE=g modisp FORCE_ERECORD=Y
Local	oracle.apps. gmo.mtl. revdisp	0	KEY	EDR_PSIG_RULE. PSIG_RULE	EDR_XML_MAP_CODE=g moundisp FORCE_ERECORD=Y
Local	oracle.apps. gmo. labelprint	2	MESSAGE	GMO_CBR_GRP. PROCESS_EVENT	N/A
Local	oracle.apps. gmo.mtl. dispense	2	KEY	GMO_CBR_GRP. PROCESS_EVENT	N/A
Local	oracle.apps. gmo.mtl. revdisp	2	KEY	GMO_CBR_GRP. PROCESS_EVENT	N/A
Local	oracle.apps. gmo.instrset. update	0	KEY	GMO_CBR_GRP. PROCESS_INSTRANCE_INSTR_SET	GMO_XML_MAP_CODE=gmoinsup

Note: Priorities are all NORMAL.

AME Events

TRANSACTION_TYPE_ID	TRANSACTION_TYPE_DESCRIPTION	LINE_ITEM_ID_QUERY
oracle.apps.gmo.dispense.setup.update	GMO ERES Dispense Setup Update	N/A
oracle.apps.gmo.mtl.revdisp	GMO ERES Material Reverse Dispense	N/A
oracle.apps.gmo.labelprint	GMO ERES Label Printing	N/A
oracle.apps.gmo.mbr.approve	GMO ERES Master Batch Record Approval	N/A
oracle.apps.gmo.instruction.complete	GMO ERES Instruction Complete	N/A
oracle.apps.gmo.dispense.setup.create	GMO ERES Dispense Setup Creation	N/A
oracle.apps.gmo.mtl.dispense	GMO ERES Material Dispense	N/A
oracle.apps.gmo.cbr.approve	GMO ERES Control Batch Record Approval	N/A

Glossary

21 CFR Part 11

FDA regulations that set forth the criteria under which the agency considers electronic records, electronic signatures, and handwritten signatures executed to electronic records to be trustworthy, reliable, and generally equivalent to paper records and handwritten signatures executed on paper.

Electronic Batch Record (EBR)

Electronic Batch Record (EBR) replaces manual procedures and paper batch records with an electronic procedure management and record-keeping system. This system is complete with electronic signatures and audit trails. An EBR, as defined by the FDA, includes text, graphics, data, audio, pictorial, or other information represented in digital form that is created, modified, maintained, archived, retrieved, or distributed by a computer system. With an EBR, all batch-relevant data are stored in the computer system.

Electronic Record (e-record)

Any combination of text, graphics, data, audio, pictorial, or other information in digital form that is created, modified, maintained, archived, retrieved, or distributed by a computer system.

Electronic Signature (e-signature)

A computer data compilation of any symbol or series of symbols executed, adopted, or authorized by an individual to be legally binding equivalent of the individual's handwritten signature.

GMO

Oracle Manufacturing Execution System (MES) for Process Manufacturing.

Gross Weight

The total weight of material, without deduction for tare, tret, or waste, as distinguished from neat, or net weight.

Label Format

A single design of a label that may include fixed data such as field prompts and variable data such as a part number. A label format is associated to a label type; it includes a subset of the available data elements from a particular label type. Multiple label formats can be associated to one label type.

Label Print Business Flow

An identifiable business process that occurs as part of material handling within Oracle Warehouse Management System. Oracle recognizes a predetermined list of business flows that can be set up to include label printing as part of the transaction processing.

Label Type

A broad category of labels that specifies the super-set of available data elements that can be included on a single label. Currently, there are eleven label types, such as Material, Serial, License Plate Number (LPN), LPN Contents, LPN Summary, Location, Shipping, Shipping Contents, and WIP Contents.

All label types are available to a WMS user. Most label types are available to an MSCA user. For instance, MSCA has no association to a license plate, so the LPN, LPN Contents, and LPN Summary label types are not applicable and therefore not available in MSCA. Multiple label formats can be defined per label type.

For example, the Material label type includes information such as Part Number, Description, Unit of Measure, Quantity, Hazard Code, Lot Number, Lot Expiration Date, Lot Attributes, and so on.

Instruction

Instructions are used to amplify the manufacturing process flow by providing details on how to execute certain steps. They also assure a documented, repeatable process; provide for ease of assembly; and improve productivity, quality, and consistency.

Net Weight

The weight of actual goods, excluding packing materials; also called actual net weight.

Tare

The weight of a container or wrapper deducted from the gross weight to obtain net weight.

WMS

Warehouse Management System product in Oracle Applications E-Business Suite.

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